

# Characterization of AISI 316L powders used in LBPF



# Index

|                      |                                                     |    |                          |            |    |
|----------------------|-----------------------------------------------------|----|--------------------------|------------|----|
| 01                   | Samples                                             | 03 | SEM vs PSD               |            |    |
| 02                   | Introduction                                        | 07 | Particle analysis        |            |    |
|                      | Powder transformations induced by LPBF process      |    | Surface oxidation        |            |    |
|                      | Hall flow test                                      |    | Powder particle porosity |            |    |
|                      | Loose density                                       |    | Build microstructure     |            |    |
|                      | Tapped density & Bulk density                       |    |                          |            |    |
|                      | Granulometry                                        |    |                          |            |    |
| 03                   | Scanning electron microscopy (SEM)                  | 20 | 04                       | Conclusion | 32 |
|                      | Results & Discussion                                |    | 05                       | References | 34 |
|                      | Powder transformations induced by LPBF process      |    | 06                       | Annexes    | 39 |
|                      | Hall flow test vs Hausner ratio                     |    |                          |            |    |
|                      | PSD vs Hausner raito                                |    |                          |            |    |
|                      | Particle size distribution of the different samples |    |                          |            |    |
|                      | PSD – Powder which has flown through the system 60x |    |                          |            |    |
| Fine powder disposal |                                                     |    |                          |            |    |

# Keywords & Abbreviations

- AISI 316L austenitic stainless steel
- Laser Powder Bed Fusion (LPBF)
- Flowability
- Hall flow test ( $F_H$ )
- Bulk density ( $\rho_b$ )
- Loose density ( $\rho_l$ )
- Tapped density ( $\rho_t$ )
- Hausner ratio ( $H_r$ )
- Compressive index ( $C_i$ )
- Laser diffraction (LD)
- Particle size distribution (PSD)
- Scanning electron microscope (SEM)
- Energy dispersive X-ray spectroscopy (EDS)

# 01 Samples

# Chemical composition given by the producer

| Elemento | C     | Si   | Mn   | Cr        | Ni        | Mo      | Fe      |
|----------|-------|------|------|-----------|-----------|---------|---------|
| Wt%      | <0.03 | <1.0 | <2.0 | 16.0-18.0 | 10.5-14.0 | 2.0-3.0 | Balance |

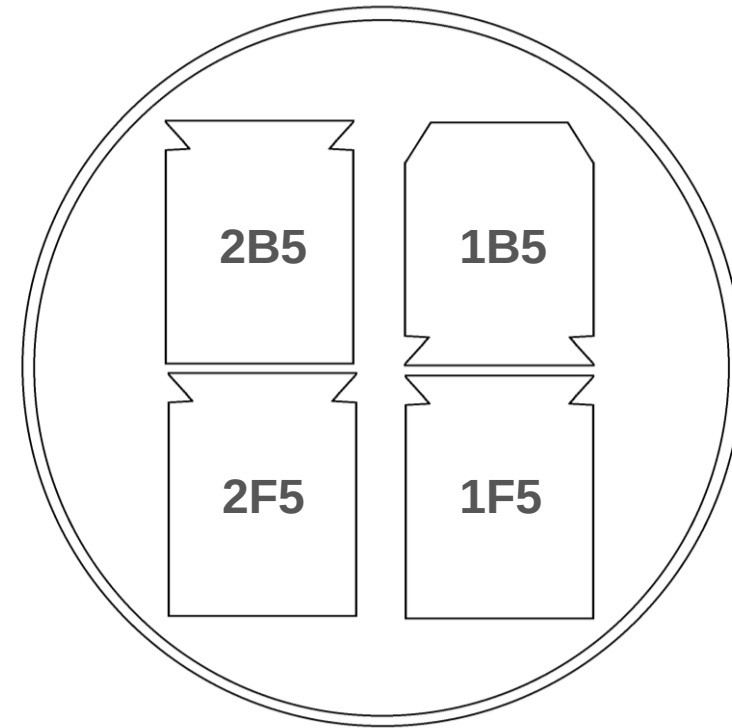
Austenitic stainless steel  
AISI 316L

# Powder Samples

- N – New AISI 316L powder
- A – Powder which has been stored and used for 2 builds
- R – Powder which has flown through the system 60x
- C – Rejected Powder
- S – Powder collected by the sieve
- F – Powder collected by the filter

# Builds Samples

- 1B5
- 1F5
- 2B5
- 2F5



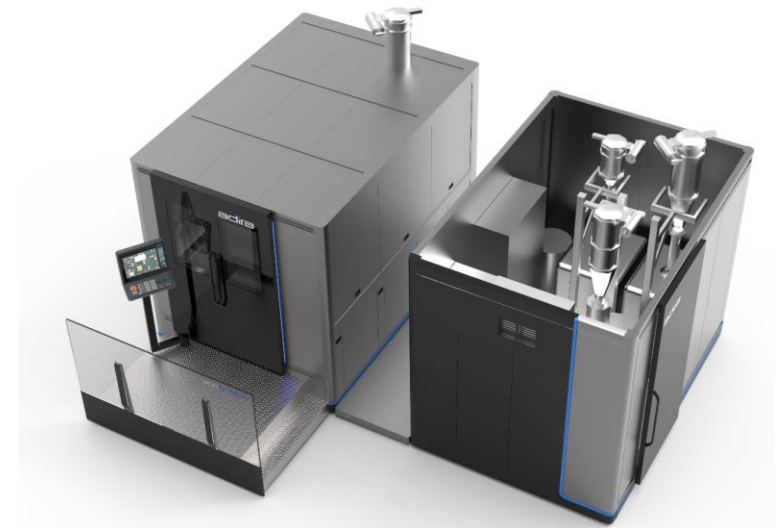
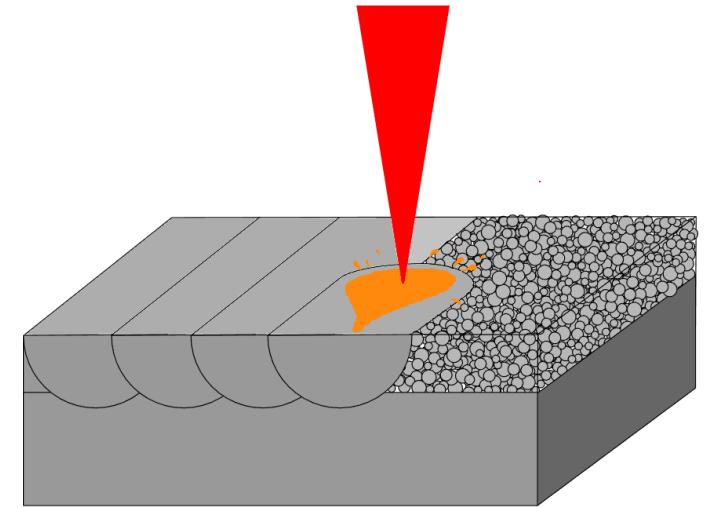
# 02 Introduction



# Introduction

Laser Powder Bed Fusion (LPBF) is an additive manufacturing technology based in a layer-by-layer approach to produce 3D objects. As the name suggests, the layers are formed by selectively melting the powder bed with the laser as shown on the figure [1, 2].

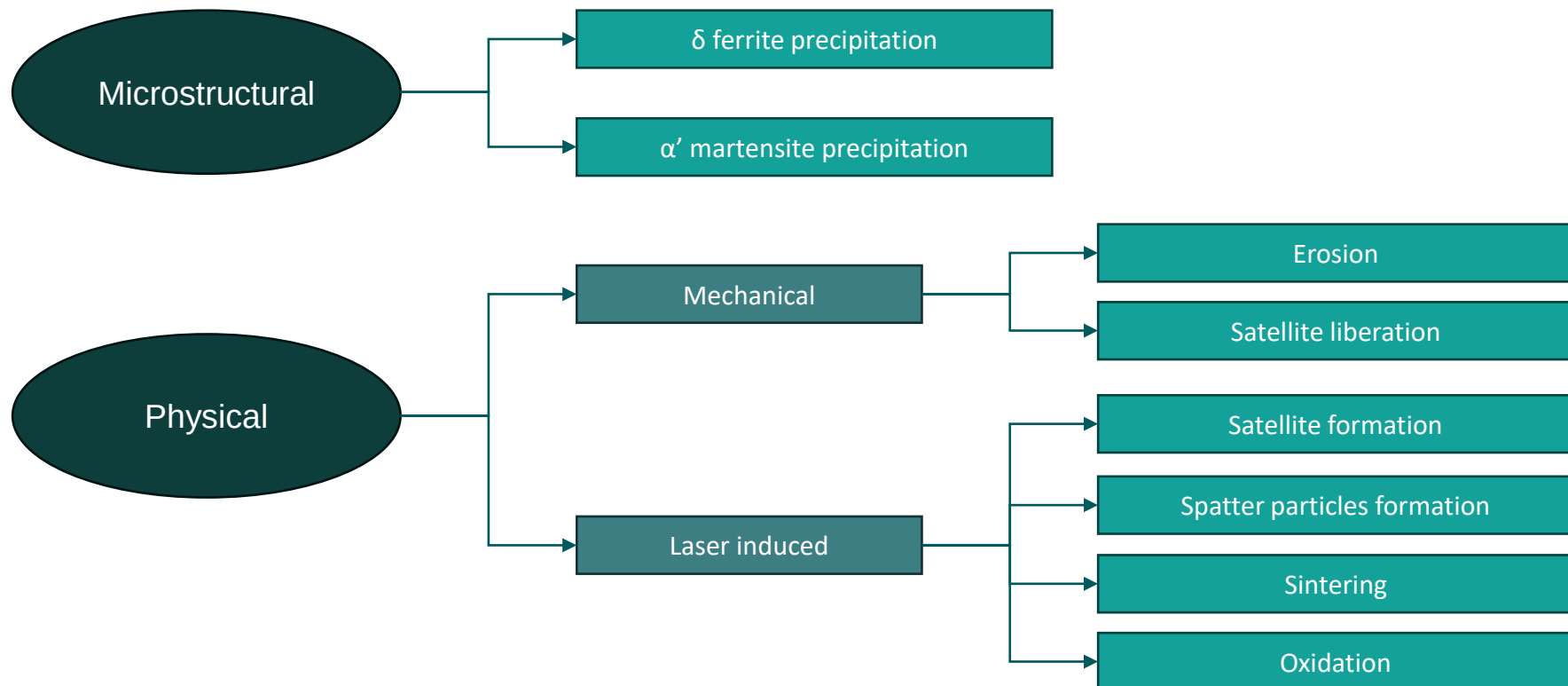
During the manufacturing process only a portion of the powder is utilized, due to the high price of feedstock and sustainability of the LPBF process, it is essential to recycle the powder [3].



# Powder transformations induced by LPBF process

During the LPBF process the properties of the stainless-steel powder are altered due to a plethora of phenomena. Powders used for LPBF need to be up to standard in many regards such as flowability and PSD. This metrics guarantee the adequate formation of homogeneous and compact layers which results in high quality builds. As such it is essential to study these phenomena to better understand powder degradation.

In a rudimentary way, powder transformations can be divided into microstructural and physical, which this last one can be further subdivided into mechanical, and laser induced phenomena.

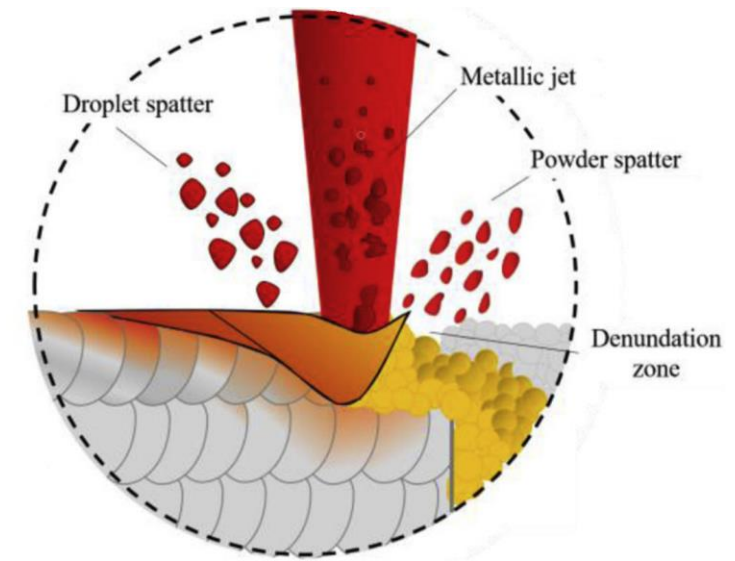


## Microstructural transformations

Although virgin AISI 316L powder has a fully austenitic ( $\gamma$ ) microstructure, during the LPBF process ferrite ( $\delta$ ) precipitation occurs, as the result of the influence of thermal cycles, which powder particles adjacent to the melt pool are exposed and the introduction of spatter particles, which due to a high cooling rate, possess a ferritic ( $\delta$ ) microstructure [4, 5].

In addition to this, austenite ( $\gamma$ ) can transform into martensite ( $\alpha'$ ) by mechanical means [6, 7]. Although there aren't any articles analysing this phenomenon in powders, it can become prevalent factor in high erosion environments where mechanical deformation can accumulate, allowing austenite ( $\gamma$ ) to transform into martensite ( $\alpha'$ ).

In contrast to austenite ( $\gamma$ ), which is a face centred cubic structure, ferrite ( $\delta$ ) and martensite ( $\alpha'$ ) are a body centred cubic structure which means they are ferromagnetic whereas austenite is paramagnetic. This is an important factor in flowability since ferromagnetic interactions worsen this metric [5, 8].

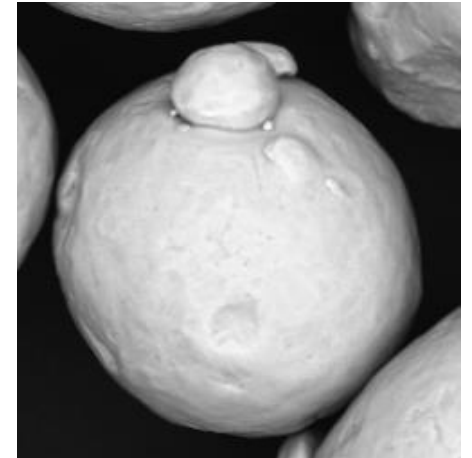


## Physical transformations of mechanical origin

During the transport of powders, the laying of the powder bed, sieving and other movements, powder particles collide with each other and other objects. These collisions erode the surface of the particles, forming small particles.

Furthermore, collisions can free satellite particles such as those seen in the figure [9]. These particles are originally formed in the atomizing process when droplets collide with each other or with already solidified particles [10].

Satellite particles have smaller dimensions than the rest of the powder and as such their liberation increases the number of finer particles present in the powder.

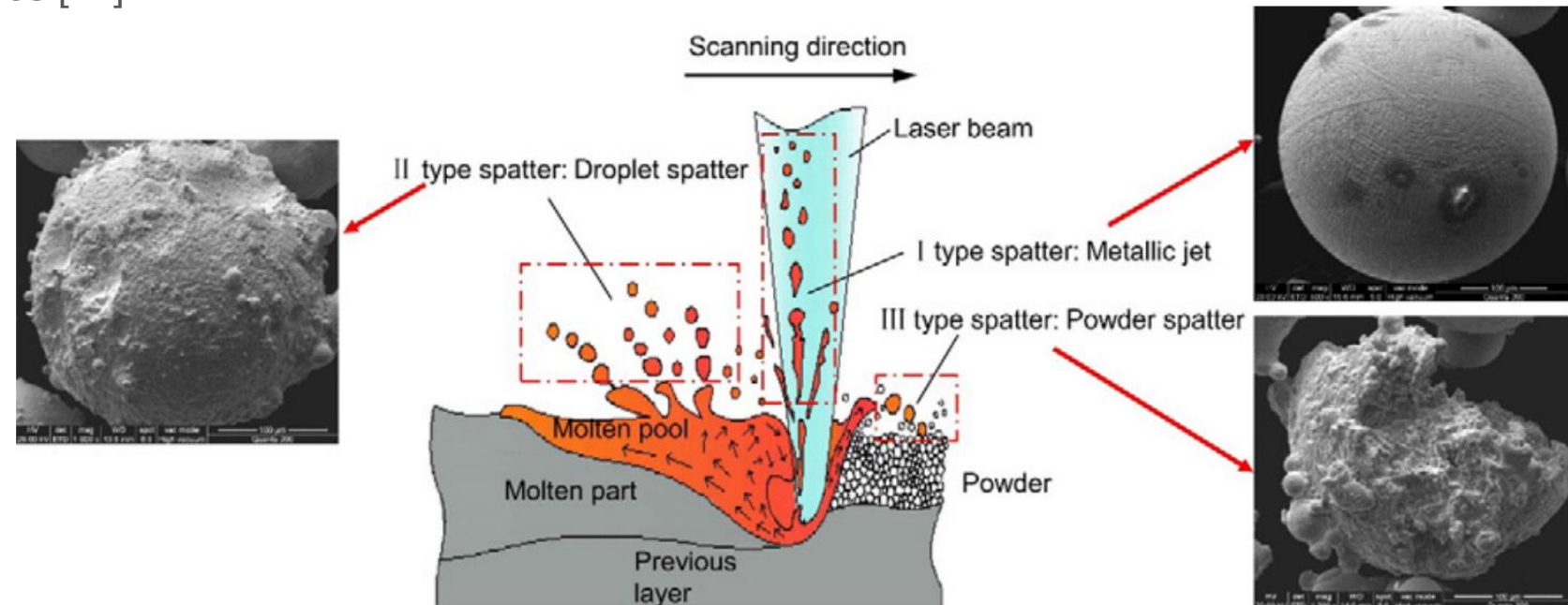


## Physical transformations induced by the laser

Satellite particles can also be formed during the LPBF process when metal vapors condense in the surface of other particles adjacent to the melt pool [11]. This phenomenon might not be as prevalent depending on the filtering system.

In addition to the formation of satellite particles, spatter particles are also formed during the LPBF process. These can be divided into type I, II & III. Type I spatter particles, also known as metallic jet, are formed when melt pool particles get ejected through the laser beam, possessing a spherical morphology. Whereas type II, also known as droplets, are formed when melt pool particles are ejected to the back of the laser beam due to the currents present in the melt pool, possessing a rough surface. Finally, type III spatter are formed when melt pool particles are ejected to the front of the laser beam, possessing an irregular morphology with satellite particles [12].

Spatter particles have greater dimensions than the virgin powder, as such they increase the number of coarse particles. Most of these particles are removed in the sieving process [13, 14].

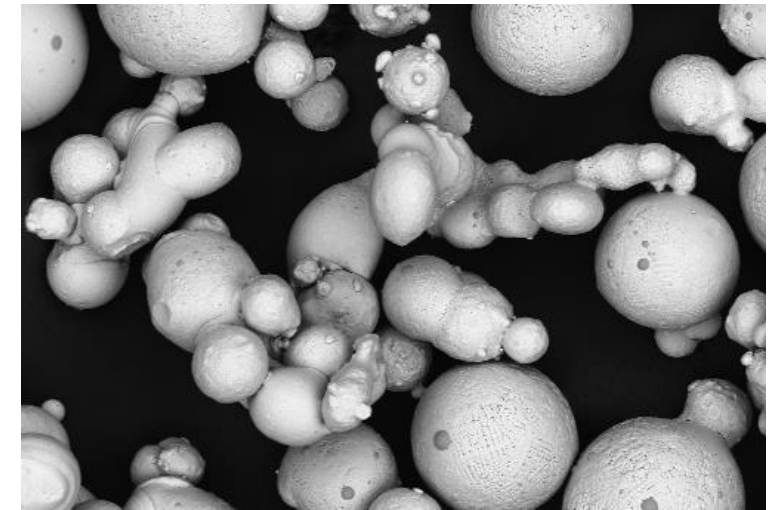


## Physical transformations induced by the laser

The introduction and subsequent increase of spatter particles leads to an increase in particles with surface oxidation, which according to the literature, are mainly composed of manganese and silicon oxide [5, 12, 13, 15].

Oxidation can also occur in the vicinity of the melt pool, altering the surface's colour. These oxides have a positive impact on the flowability of reused powders, as they reduce the friction between particles [13, 16].

Sintered particles or agglomerates formed from semi-liquid spattered particle which fall on the powder bed are mostly removed by sieving. These agglomerates preferentially forms from smaller particles [13].

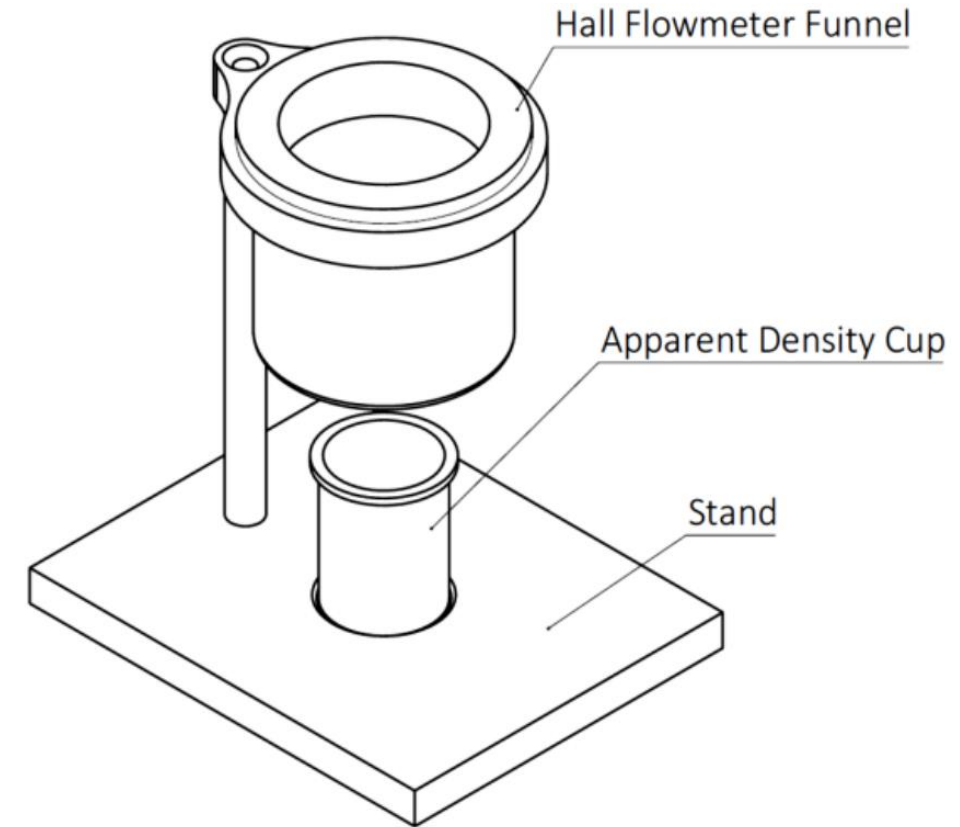


## Hall flow test

The Hall flow test is a flow rate method of measuring powder flowability. This method consists in measuring the time it takes for 50 g of powder to flow through the Hall funnel. The shorter the flow time, the better the flowability [17].

According to the LPW equipment's instructions, the values obtained must be multiplied by a corrective factor of 0.92 to account for human error.

The [ASTM F3049-14](#) standard recommends flow rate methods such as Hall funnel, Carney funnel or Arnold meter to assess the flowability of powders used for additive manufacturing. This is because these methods better resemble the movement of powder particles during the process [17, 18].



**Figure 1 Hall Flow test set-up**



## Loose density

Loose density is the density resulting from the subtle stacking of 316L powder grains after free-falling through the [Copley Scott Volumeter](#).

In the test, the powders are poured into a device which controls the falling of the powder's particles until they completely fill a 25.00 ml container. The excess is then removed with a spatula, in a careful manner not to compact the powders. Lastly, the weight of the container is subtracted from the total weigh of the container full of powder and the loose density is calculated.

$$m_{\text{powder}} = (m_{\text{container with powder}} - m_{\text{container}})$$

$$\rho_{\text{Loose}} = \frac{m_{\text{powder}}}{25 \text{ ml}}$$



Copley Bulk Density Tester  
(Scott Volumeter)



# Tapped density & Bulk density

Bulk density is similar to Loose density apart from the falling and measurement method. When measuring Bulk density, 300 grams of powder are poured directly into the apparatus and the volume is measured by averaging the lower volume (V1) and the upper volume (V2) and calculating the density as seen in the figure [19].

After measuring the Bulk density, the Tapped density test begins. At the end of 750 strokes, the volume is measured using the same method and the Tapped density is calculated.

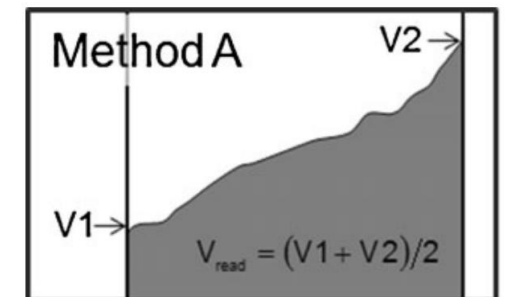
The Bulk & Tapped density tests were carried out on a [Copley JV 100i](#) and did not comply with the standard nom. The tests used 300 g of powder with 250 taps per minute (taps/min) for 3 minutes.

Using the Bulk & Tapped density values, it is possible to calculate the Hausner ratio ( $H_r$ ) and the Compressive index ( $C_i$ ). These values are a measurement of flowability by density ratio, the higher the Hausner ratio, the worse the flowability of the powder [17].

The  $H_r$  &  $C_i$  represent the ability of powders to compact, being an indicator of flowability since the particles need to move to increase density [17].

$$H_r = \frac{\rho_{Bulk}}{\rho_{Tapped}}$$

$$C_i = \frac{(\rho_{Bulk} - \rho_{Tapped})}{\rho_{Tapped}}$$

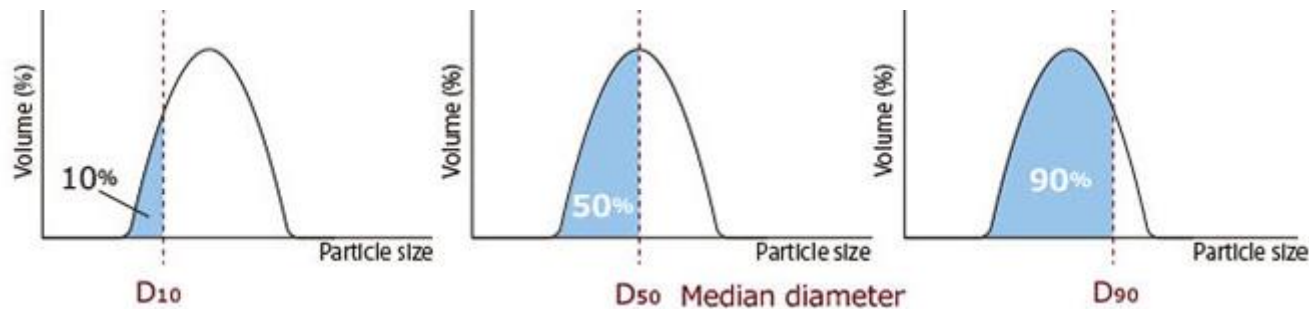


# Granulometry

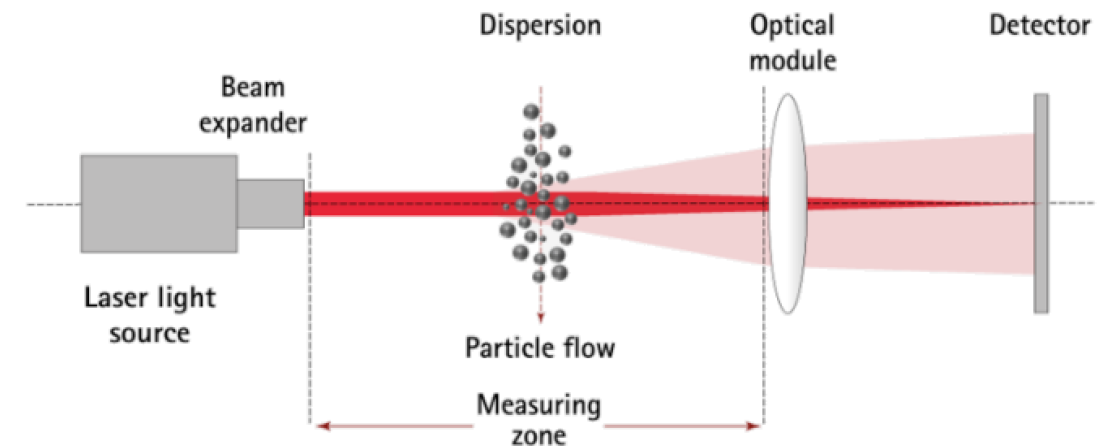
Granulometry is the study of the size distribution of particles in a powder. The results were obtained by laser diffraction using a Beckman Coulter LS230 Laser Diffraction Particle Size Analyzer available at DEQ, FEUP.

In laser diffraction a stream of a suspension with powder particles passes through a laser producing a diffraction pattern, with this pattern the volume percentage and size is determine, plotting a graph volume percentage vs size [20].

This analysis, in addition to the graph, provides us  $D_{10}$ ,  $D_{50}$  &  $D_{90}$  values which correspond to the powder particle size corresponding to the accumulative percentage of 10%, 50% & 90%. Using this data, it is possible to calculate the Span ( $\text{Span} = \frac{D_{90} - D_{10}}{D_{50}}$ ) which is used to assess the distribution of the powders. The smaller the span is the more uniform the size of the powder particles [20].



Beckman Coulter LS230 Laser Diffraction Particle Size Analyzer

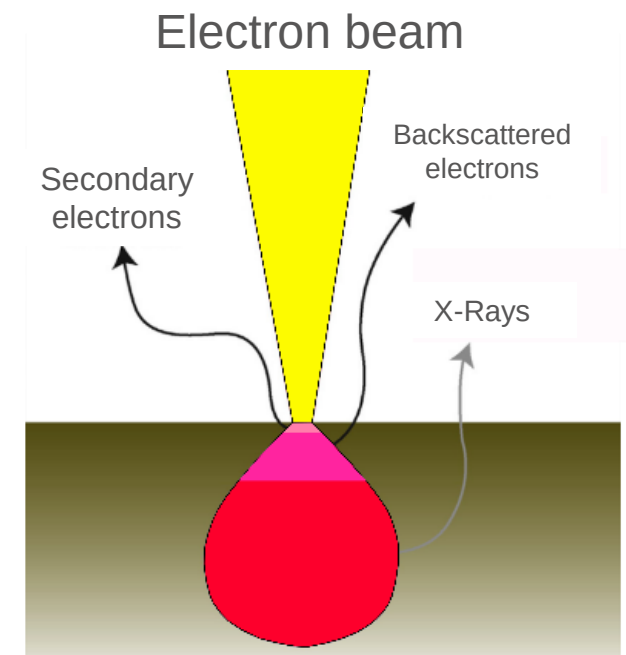
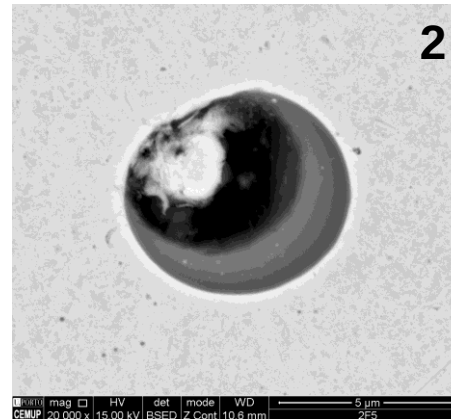
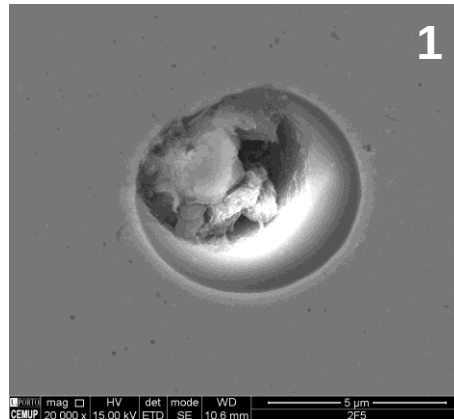


# Scanning Electron Microscopy (SEM)

It is not possible to conclude on the morphology of powder particles through PSD, it is therefore essential to complement this analysis with scanning electron microscopy (SEM) to qualitatively assess the morphology of the particles [16].

SEM, unlike optical microscopy which uses a beam of photons to obtain images, uses an electron beam to scan the surface of the samples and obtain an image [21].

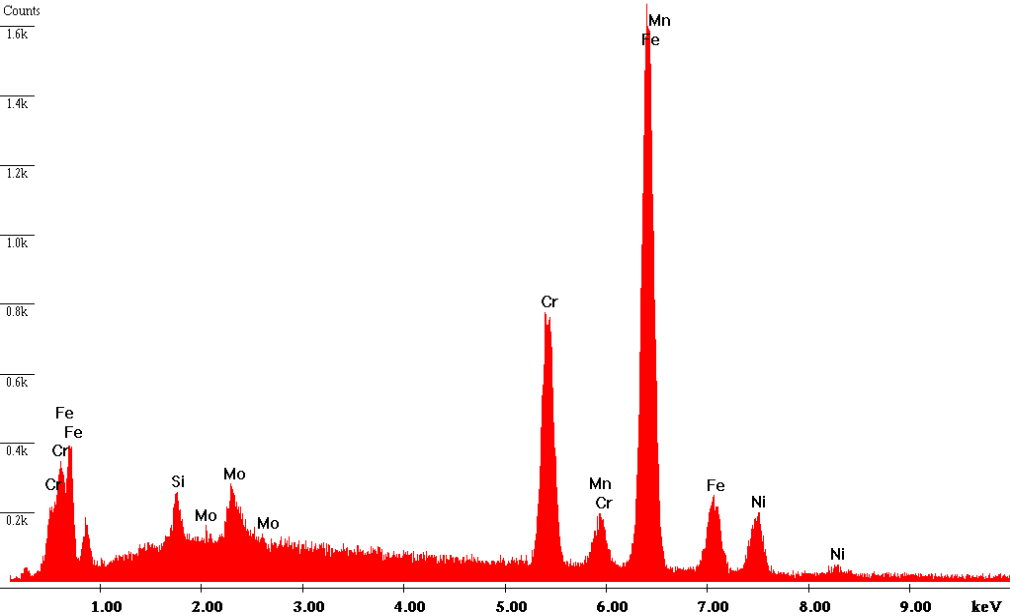
There are two different types of electrons that are used in SEM, secondary electrons, and backscattered electrons. Secondary electrons provide a topological contrast to the images (1) unlike backscattered electrons which provide an atomic number contrast (2), the higher the atomic number of the element the brighter it will appear [21].



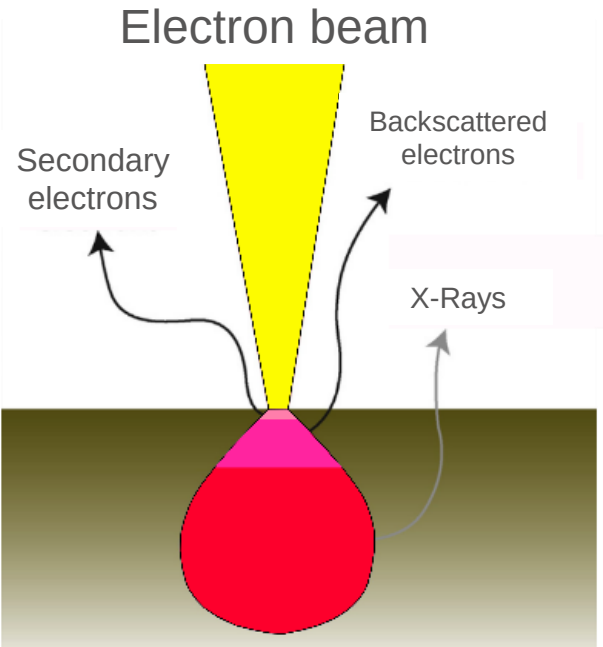
# Scanning Electron Microscopy (SEM)

SEM also has the capability to analyse the chemical composition through energy dispersive X-ray spectroscopy (EDS). Something to bear in mind with this analysis is the existence of an interaction zone. The chemical composition obtained corresponds to the chemical composition of the interaction zone, and this factor must be considered when interpreting the results [22].

Label A: CEMUP 15keV N Z1



| Elemento | Wt%   | At%   |
|----------|-------|-------|
| Si       | 1.02  | 2.03  |
| Mo       | 2.98  | 1.73  |
| Cr       | 16.77 | 17.92 |
| Mn       | 2.12  | 2.14  |
| Fe       | 65.22 | 64.92 |
| Ni       | 11.89 | 11.25 |

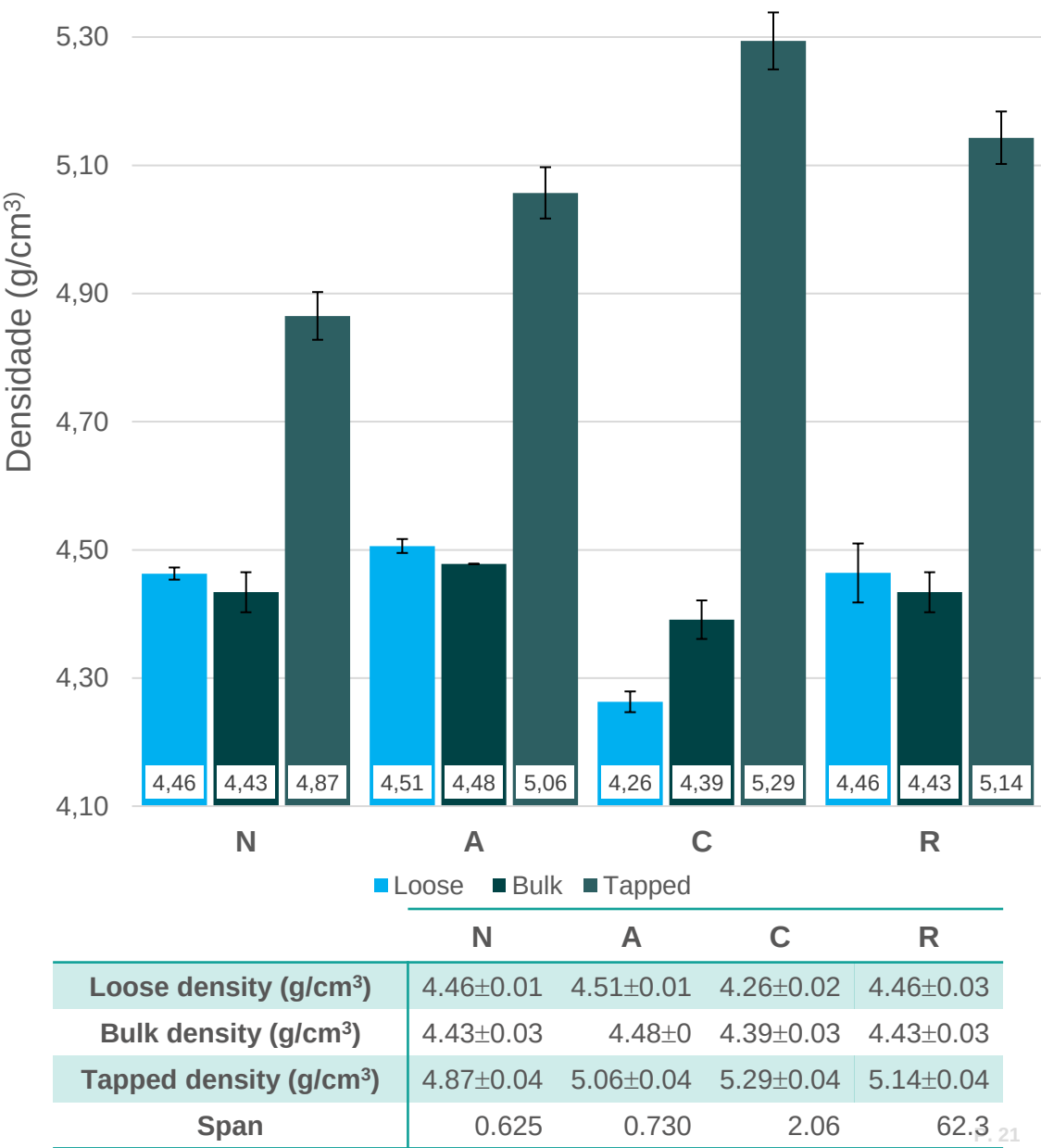


# 03 Results & Discussion

# Loose, Bulk & Tapped density

The values of Bulk and Loose density are similar between all the sampled powders except for the powder that has been used in 2 builds (A), which as seen an increase compared to the other samples. This can be attributed to the increase of larger span & coarser particles and the surface oxidation [13].

The circulation of powder through the system 60 times (R) showed an identical bulk and loose density to virgin powder, despite a significant increase of span, which is indicative of the difficulty of movement that particles have in the powder. This also happens in the rejected powders (C) and can be partially attributed to their bimodal nature of PSD.

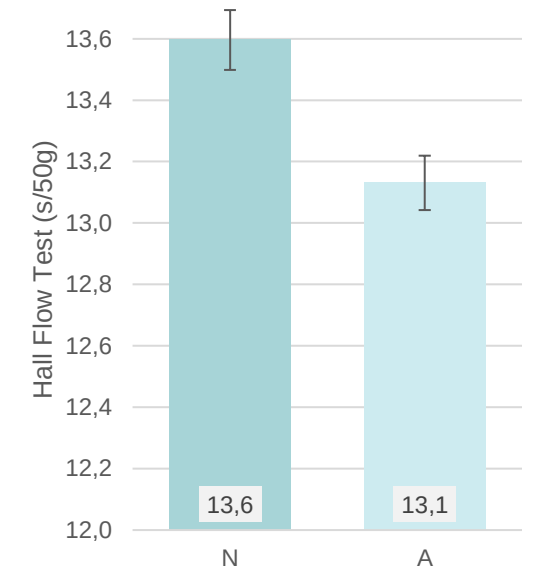
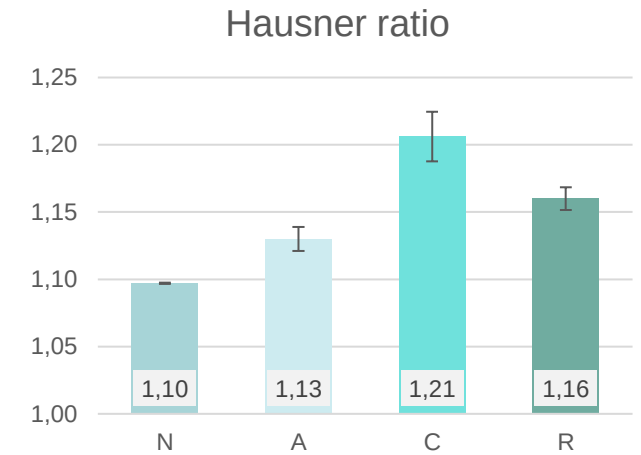


## Hall flow test vs Hausner ratio

Marchetti et al. hinted that the relationship between Hall flow test and Hausner ratio is strongly linear (Pearson coefficient  $\sim 1$ ) [17]. The rejected powder (C) and the powder which has flowed through the system 60 times (R) did not flow from the Hall funnel, so the time could be assumed to be greater (worse flowability) than that of the virgin powder (N). This partially verifies the relationship between the Hall flow test and the Hausner ratio.

However, there is a discordance. According to the Hall flow test, the powder which was used in 2 builds (A) has better flowability than the N powder. According to the Hausner ratio, the A powder have worse flowability than the N powder.

This discordance can be attributed to the fact that these tests belong to different flowability test methods. Since the Hall flow test more closely resembles the powder flow conditions in the LPBF process, it can be concluded that flowability increased after 2 builds [17, 18].



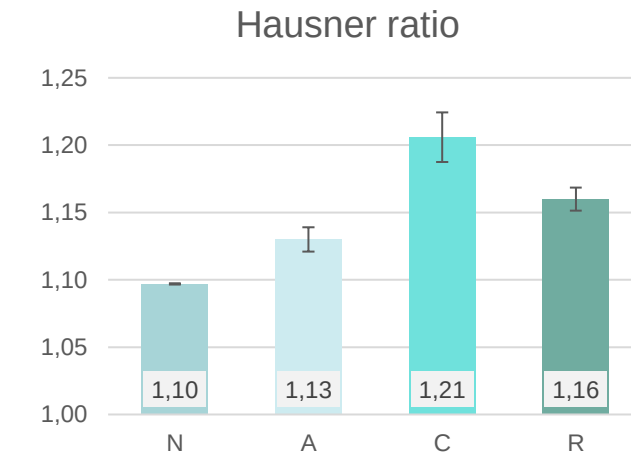
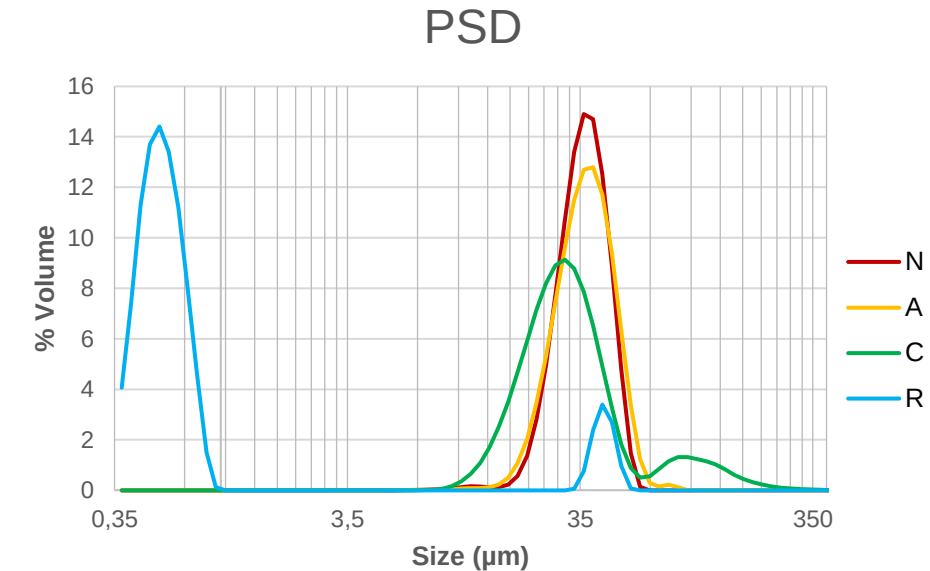
# PSD vs Hausner ratio

When analyzing the particle size distribution and the Hausner ratio, it is possible to see a relationship. This is due to the higher the Span, the greater the compatibility of the powders, so the greater the difference between Bulk/Loose density and Tapped density.

So  $0.625 < 0.730 < 2.06 \rightarrow 1.10 < 1.13 < 1.21$ . The Span value of the recirculated powders is not indicative of their flowability due to the difference between the average of the curves.

Following the same reasoning, it is to be expected that if the powders recirculated 60 times (R) pass through they will have a lower Hausner ratio than the new powders (N).

|                            | N        | A        | R        | R+    | C        |
|----------------------------|----------|----------|----------|-------|----------|
| <b>D<sub>10</sub> (μm)</b> | 27.00 μm | 25.90 μm | 0.444 μm | 36.56 | 18.84 μm |
| <b>D<sub>50</sub> (μm)</b> | 37.99 μm | 38.54 μm | 0.595 μm | 42.03 | 31.72 μm |
| <b>D<sub>90</sub> (μm)</b> | 50.73 μm | 54.04 μm | 37.51 μm | 47.99 | 84.17 μm |
| <b>Span</b>                | 0.625    | 0.730    | 62.3     | 0.272 | 2.06     |





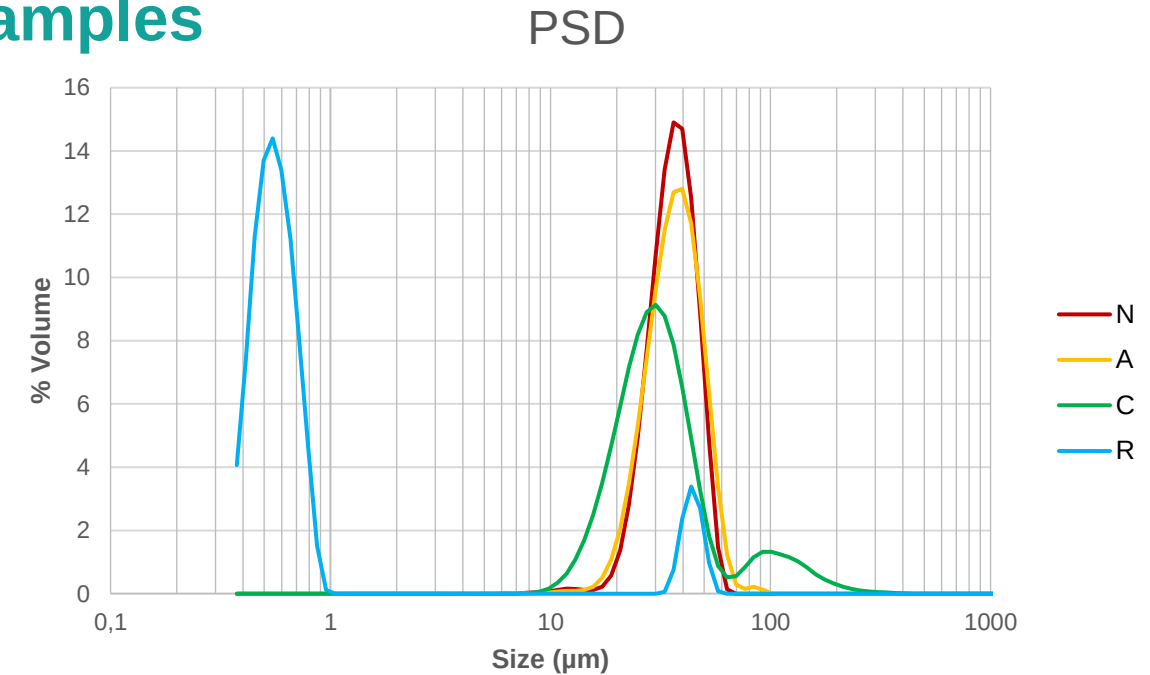
## Particle size distribution of the different samples

Comparing the new powders (N) with the powders used in 2 builds (A) shows an increase in  $D_{50}$  and  $D_{90}$ , which is in line with the literature. This increase is due to the creation of spatter particles [13].

Contrary to what most literatures reports,  $D_{10}$  decreased with recuse [14]. This may be due to the fact that the builds have high volumes and low mass, with fewer spatter particles introduced [23].

The powders recirculated 60 times (R) show two populations with different particle sizes, with the upper curve (curve with higher particle size) resembling the N & A powders.

The rejected powders (C), like the R powders, have two populations. The upper curve of these powders has the coarsest grain size of all the powders.



|               | N     | A     | R     | C     |
|---------------|-------|-------|-------|-------|
| $D_{10}$ (μm) | 27.00 | 25.90 | 0.444 | 18.84 |
| $D_{50}$ (μm) | 37.99 | 38.54 | 0.595 | 31.72 |
| $D_{90}$ (μm) | 50.73 | 54.04 | 37.51 | 84.17 |
| Span          | 0.625 | 0.730 | 62.3  | 2.06  |

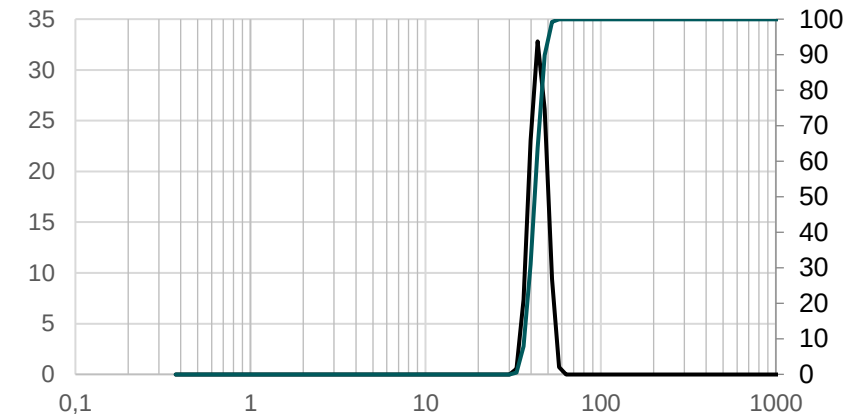
## PSD – Powder which has flown through the system 60x (R)

The high-volume percentage of fine particle shown in the PSD analysis does not match the observation made in SEM imaging. The under representation of fine particles in SEM imaging is due to the preparation of the sample, during this procedure the sample is blown to free loose particles, since fine particles have a lesser mass, they are preferentially blown.

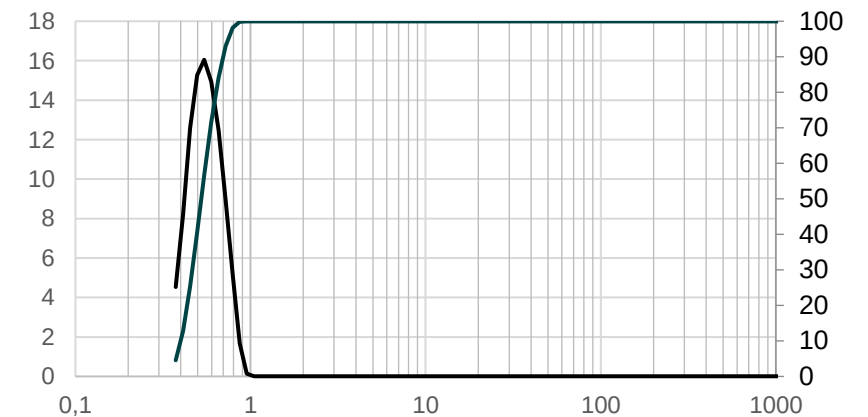
Nevertheless, the volume percentage shown in the PSD is most certainly over exaggerated. These can be due to segregation of the powder and an incorrect sampling or due a segregation in the liquid medium, finer particles stay in suspension more easily than coarser ones, this can explain a preferential sampling of finer particles which appear in the PSD analysis.

|          | Superior | Inferior |
|----------|----------|----------|
| D10 (µm) | 36.56    | 0.3996   |
| D50 (µm) | 42.03    | 0.5247   |
| D90 (µm) | 47.99    | 0.6994   |
| Span     | 0.272    | 0.571    |

R - Curva superior (10.33%)



R - Curva inferior (89.67%)



## Fine powder disposal

Powder disposal is usually done through mixing the powders within a medium creating a suspension. EGEO is a Portuguese company who specialises in residue management.



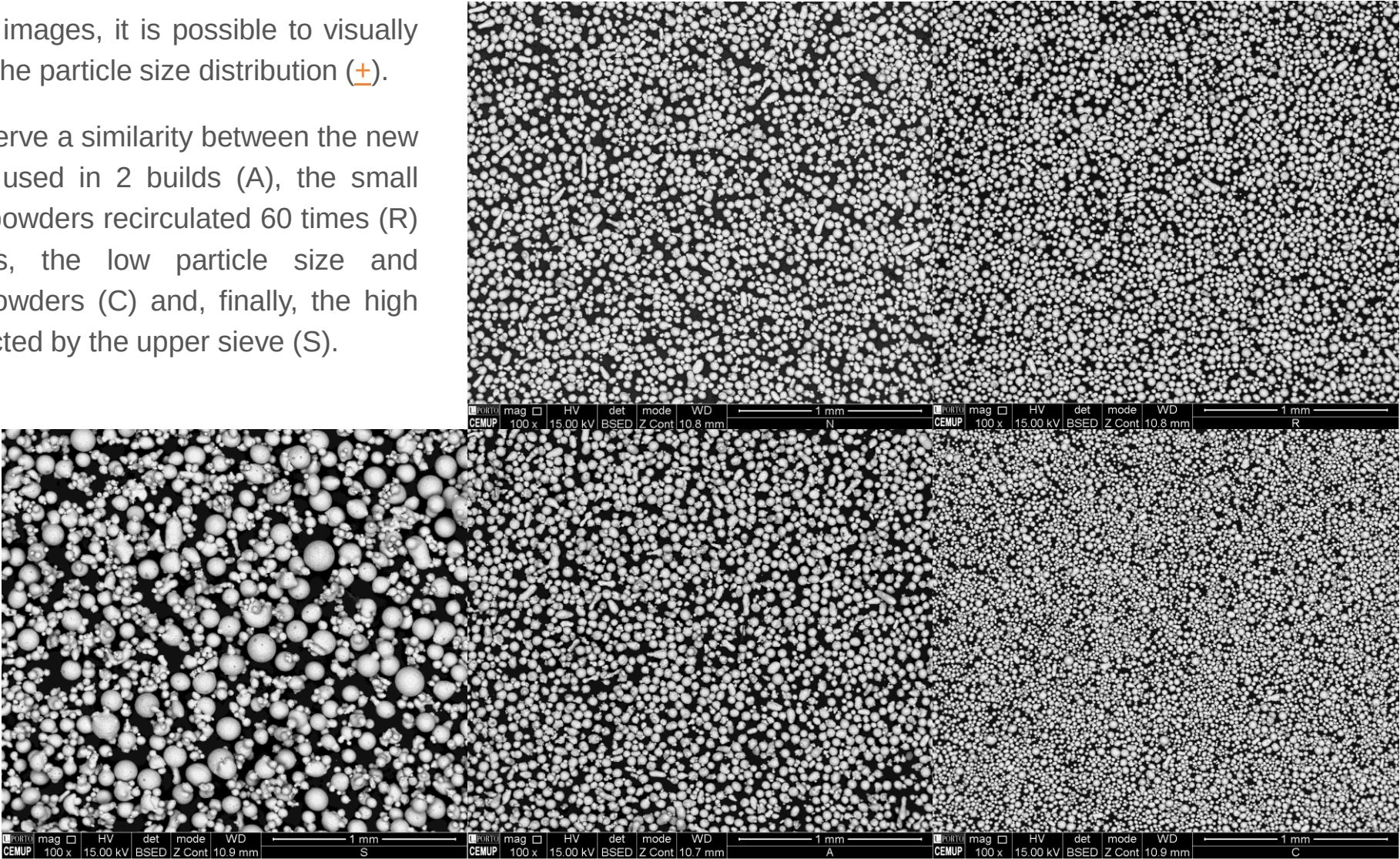


# SEM vs PSD

In the following atomic contrast images, it is possible to visually check the results obtained from the particle size distribution (+).

In particular, it is possible to observe a similarity between the new powders (N) and the powders used in 2 builds (A), the small reduction in particle size of the powders recirculated 60 times (R) compared to the N powders, the low particle size and heterogeneity of the rejected powders (C) and, finally, the high particle size of the powders rejected by the upper sieve (S).

|                      | N     | A     | R+    | C     |
|----------------------|-------|-------|-------|-------|
| D <sub>50</sub> (µm) | 37.99 | 38.54 | 42.03 | 31.72 |



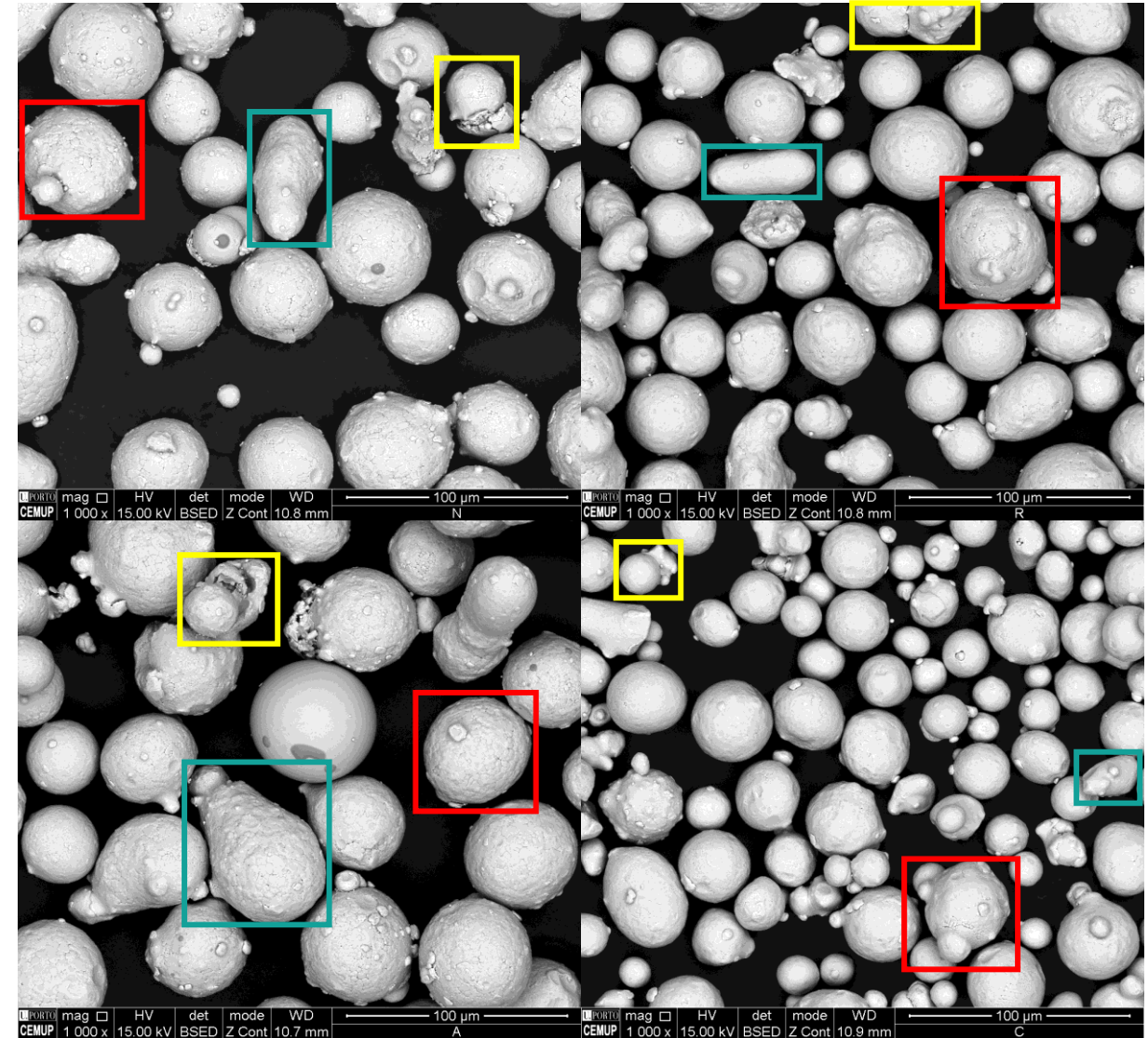


# Particle analysis

It's possible to note a variety of particles shapes and sizes in all the samples. All the samples possess elongated particles (blue), satellite particles (red) and collided particles (yellow).

A further analysis must be done to assess quantitatively the number of these particles and related them with the different histories of the samples. (+)

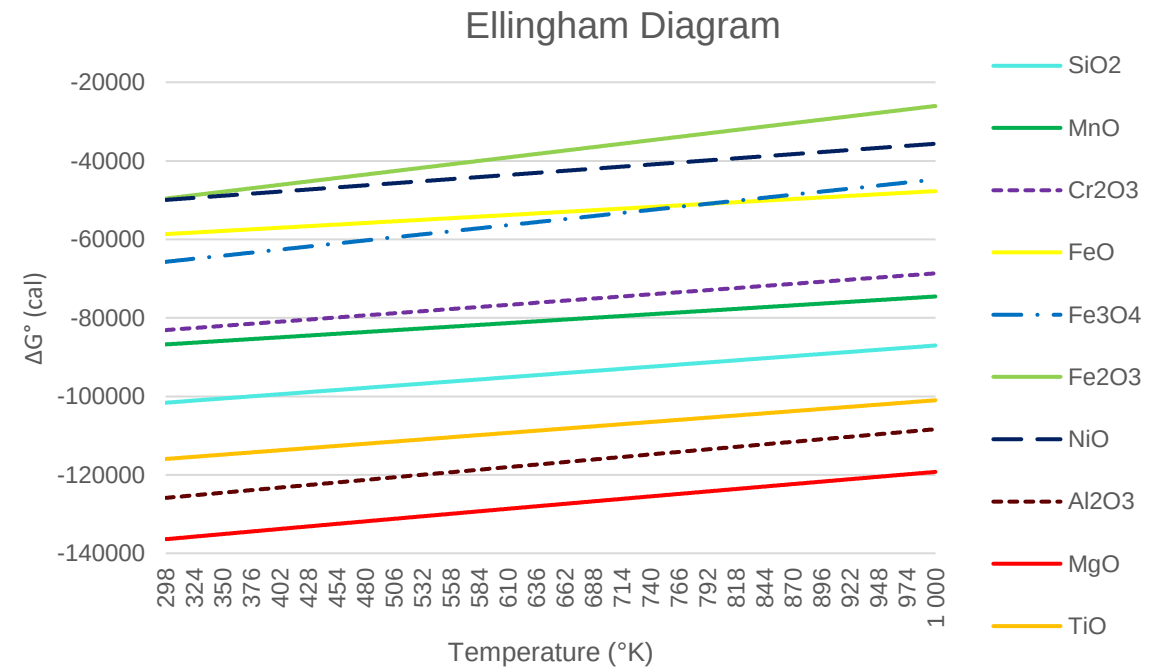
Oxide particles were also found in the virgin powder (N), this is also reported in other AISI 316L gas atomized powders [5]. (+)



# Surface oxidation

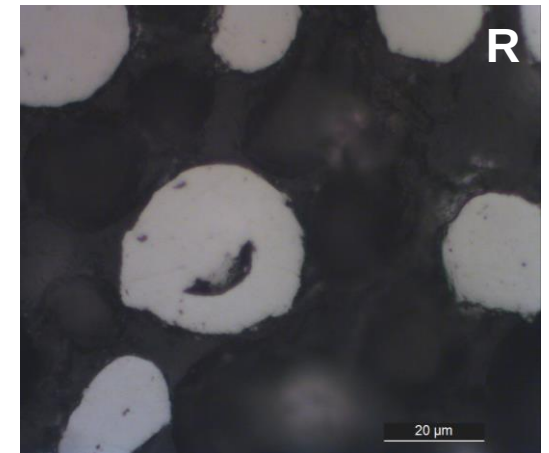
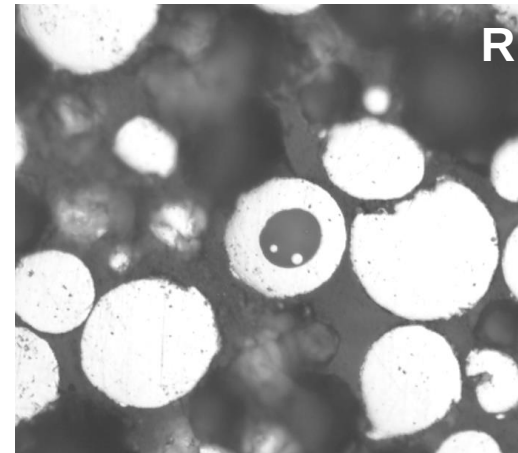
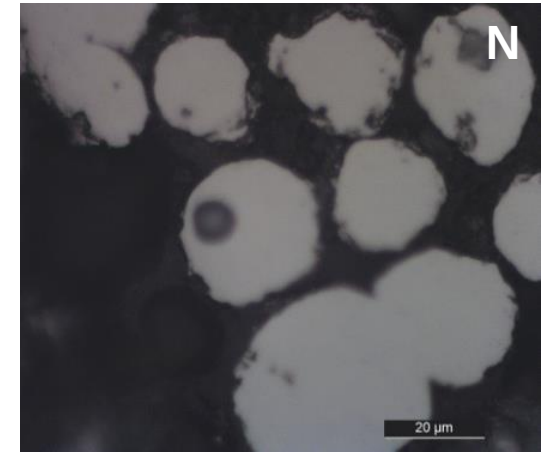
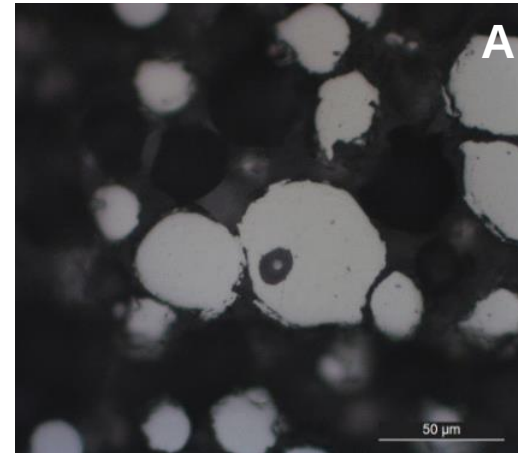
Through to Ellingham Diagram it is possible to explain the prevalence of silicon and manganese, followed by chromium and iron on the oxide nodules, since these elements have a lower Gibbs free energy to form oxides than the other elements present on the matrix.

The appearance of aluminum and magnesium on the oxide nodules can be explain due to their extremely low Gibbs free energy. Although they exist in such low percentages in the matrix that do not appear on the EDS analysis, due to their high affinity to oxygen they appear in high concentrations in the nodules. Aluminum is used in the steel making process, so it is expected to be presence in trace amounts on the steel, it also has been seen on 316L oxide in the literature [5]. (±,±)



## Powder particle porosity

Powders N, A and R presented internal porosity, but further analysis must be done to quantitatively evaluated the porosity variation in these powders, such as X-CT analysis. Powder porosity is an intrinsic quality of gas atomized powders and can vary in the process [11]. (⊕)

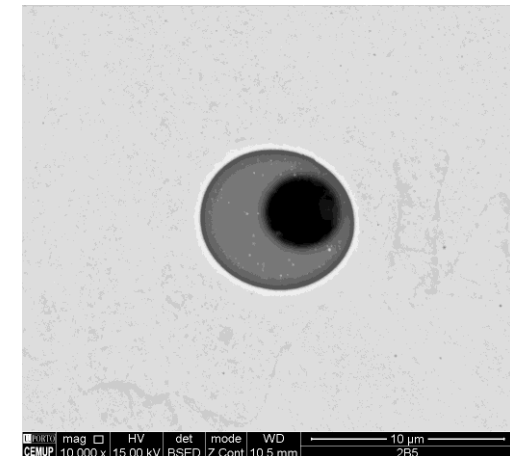
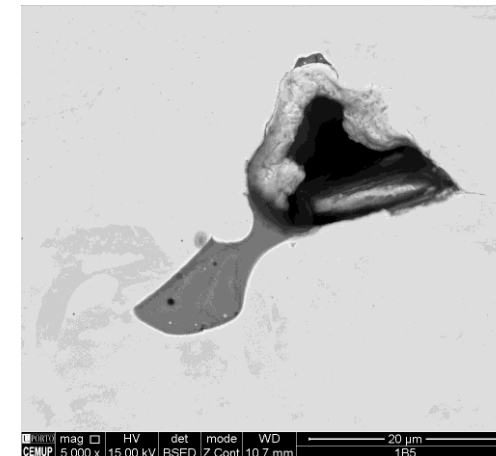
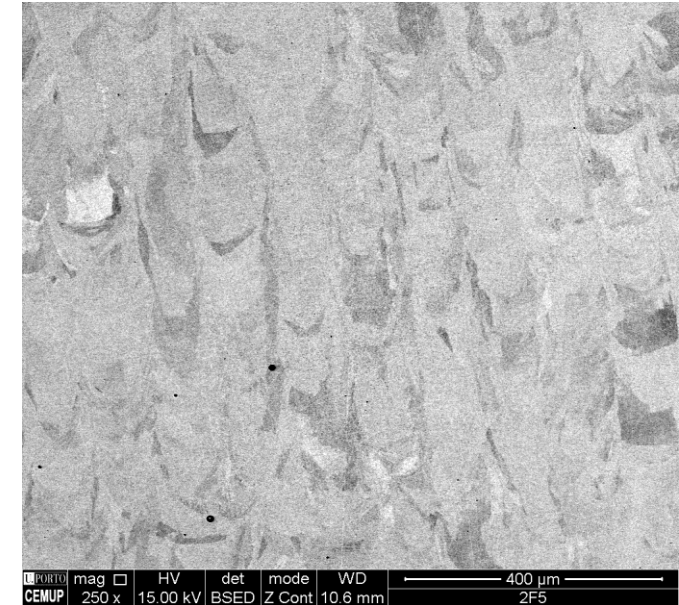


## Build microstructure

The microstructure of the builds were obtained by SEM. It's possible to note melt pools, oxide nodules and porosity. (±, ±, ±, ±)

The porosity could be characterized into two types, irregular shaped which are attributed to a lack of fusion, and spherical shaped which are usually formed due to gas entrapment [24].

The oxidation nodules present in the microstructure of the final build are similar in composition to the surface oxidation present in the powders.





# 04 Conclusion

## Conclusion

- There is an increase of flowability, span and  $D_{50}$  associated with 2 reuses of powders, as well as a reduction of the  $D_{10}$ ;
- Powders C & R have a bimodal PSD and lower flowability;
- The powder's surface oxide nodules and oxides present in the final build are main silicon and manganese oxides with smaller quantities of chromium, aluminium and iron;
- All powder samples possess oddly shaped particles and porosity;

# 05 References

# Bibliografia

- [1] – Tan, J.H., W.L.E. Wong, and K.W. Dalgarno, An overview of powder granulometry on feedstock and part performance in the selective laser melting process. Additive Manufacturing, 2017. 18: p. 228-255. <https://doi.org/10.1016/j.addma.2017.10.011>
- [2] – Chowdhury, S., et al., Laser powder bed fusion: a state-of-the-art review of the technology, materials, properties & defects, and numerical modelling. Journal of Materials Research and Technology, 2022. 20: p. 2109-2172. <https://doi.org/10.1016/j.jmrt.2022.07.121>
- [3] – Teubler, J., et al., Critical evaluation of the material characteristics and environmental potential of laser beam melting processes for the additive manufacturing of metallic components. Journal of Cleaner Production, 2019. 237. <https://doi.org/10.1016/j.jclepro.2019.117775>
- [4] – Pinto, F.C., et al., Defects in parts manufactured by selective laser melting caused by  $\delta$ -ferrite in reused 316L steel powder feedstock. Additive Manufacturing, 2020. 31. <https://doi.org/10.1016/j.addma.2019.100979>
- [5] – Heiden, M.J., et al., Evolution of 316L stainless steel feedstock due to laser powder bed fusion process. Additive Manufacturing, 2019. 25: p. 84-103. <https://doi.org/10.1016/j.addma.2018.10.019>
- [6] – Zhang, S.Y., et al., Investigation of Martensite Transformation in 316L Stainless Steel. Materials Today: Proceedings, 2015. 2: p. S251-S260. <https://doi.org/10.1016/j.matpr.2015.05.035>
- [7] – Ni, X., et al., Deformation-induced martensitic transformation in 316L stainless steels fabricated by laser powder bed fusion. Materials Letters, 2021. 302. <https://doi.org/10.1016/j.matlet.2021.130377>
- [8] – Hulme, C.N., Flow behavior of magnetic steel powder. Particulate Science and Technology, 2021. 40(5): p. 576-588. <https://doi.org/10.1080/02726351.2021.1977442>
- [9] – Tang, H.P., et al., Effect of Powder Reuse Times on Additive Manufacturing of Ti-6Al-4V by Selective Electron Beam Melting. Jom, 2015. 67(3): p. 555-563. <https://doi.org/10.1007/s11837-015-1300-4>

# Bibliografia

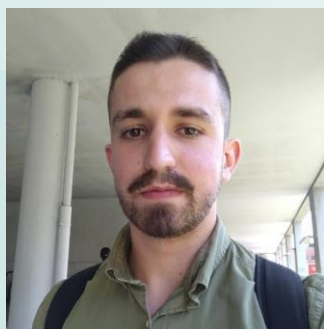
- [10] – Anderson, I.E., E.M.H. White, and R. Dehoff, Feedstock powder processing research needs for additive manufacturing development. Current Opinion in Solid State and Materials Science, 2018. 22(1): p. 8-15. <https://doi.org/10.1016/j.cossms.2018.01.002>
- [11] – Xiong, L., et al., Defect and satellite characteristics of additive manufacturing metal powders. Advanced Powder Technology, 2022. 33(3). <https://doi.org/10.1016/j.appt.2022.103486>
- [12] – Wang, D., et al., Mechanisms and characteristics of spatter generation in SLM processing and its effect on the properties. Materials & Design, 2017. 117: p. 121-130. <https://doi.org/10.1016/j.matdes.2016.12.060>
- [13] – Timothée Delacroix, F.L., Frédéric Schuster, Hicham Maskrot, Jean-Paul Garandet,, Influence of powder recycling on 316L stainless steel feedstocks and printed parts in laser powder bed fusion. Additive Manufacturing,, 2022. Volume 50,, <https://doi.org/https://doi.org/10.1016/j.addma.2021.102553>
- [14] – Gorji, N.E., et al., Recyclability of stainless steel (316 L) powder within the additive manufacturing process. Materialia, 2019. 8. <https://doi.org/10.1016/j.mtla.2019.100489>
- [15] – Simonelli, M., et al., A Study on the Laser Spatter and the Oxidation Reactions During Selective Laser Melting of 316L Stainless Steel, Al-Si10-Mg, and Ti-6Al-4V. Metallurgical and Materials Transactions A, 2015. 46(9): p. 3842-3851. <https://doi.org/10.1007/s11661-015-2882-8>
- [16] – Sutton, A.T., et al., Characterization of AISI 304L stainless steel powder recycled in the laser powder-bed fusion process. Additive Manufacturing, 2020. 32. <https://doi.org/10.1016/j.addma.2019.100981>
- [17] – Marchetti, L. and C. Hulme-Smith, Flowability of steel and tool steel powders: A comparison between testing methods. Powder Technology, 2021. 384: p. 402-413. <https://doi.org/10.1016/j.powtec.2021.01.074>
- [18] – ASTM, F3049-14, in Standard Guide for Characterizing Properties of Metal Powders Used for Additive Manufacturing Processes. 2021. p. 3.

# Bibliografia

- [19] – Sousa e Silva, J.P., et al., Note on the measurement of bulk density and tapped density of powders according to the European Pharmacopeia. AAPS PharmSciTech, 2013. 14(3): p. 1098-100. <https://doi.org/10.1208/s12249-013-9994-5>
- [20] – Grubbs, J., et al., Comparison of laser diffraction and image analysis techniques for particle size-shape characterization in additive manufacturing applications. Powder Technology, 2021. 391: p. 20-33. <https://doi.org/10.1016/j.powtec.2021.06.003>
- [21] – Nanakoudis, A. What is SEM? Scanning Electron Microscopy Explained. 2023 25/07/2023]; Available from: <https://www.thermofisher.com/blog/materials/what-is-sem-scanning-electron-microscopy-explained/>.
- [22] – Nanakoudis, A. EDX Analysis with SEM: How Does it Work? 2019 28/11/2019 [cited 2023 25/07/2023]; Available from: <https://www.thermofisher.com/blog/materials/edx-analysis-with-sem-how-does-it-work/>.
- [23] – Kriewall, C.S., et al., Effects of Area Fraction and Part Spacing on Degradation of 304L Stainless Steel Powder in Selective Laser Melting. Solid Freeform Fabrication 2017: Proceedings of the 28th Annual International, 2017. <https://hdl.handle.net/2152/89820>
- [24] – Hosseini, E. and V.A. Popovich, A review of mechanical properties of additively manufactured Inconel 718. Additive Manufacturing, 2019. 30. <https://doi.org/10.1016/j.addma.2019.100877>

# Contactos

## Equipa



### **Ilídio Costa**

Estudante de LEMAT

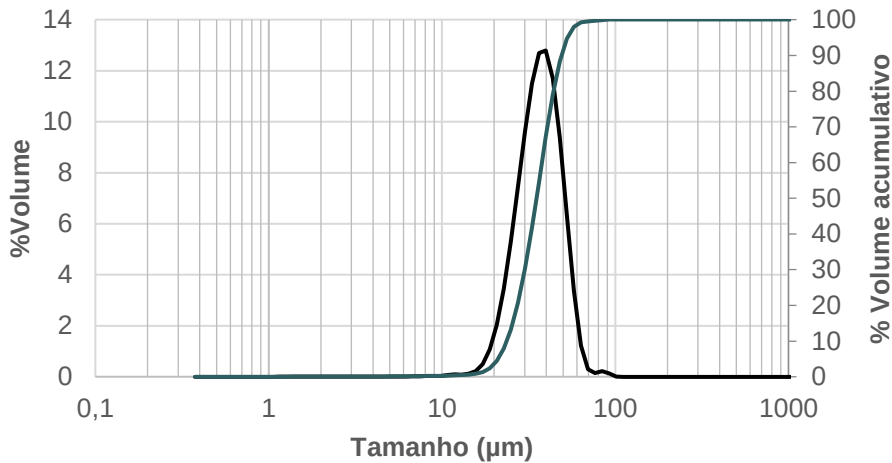
T. +351 925870614

E. [ilidiomibritocosta@gmail.com](mailto:ilidiomibritocosta@gmail.com)

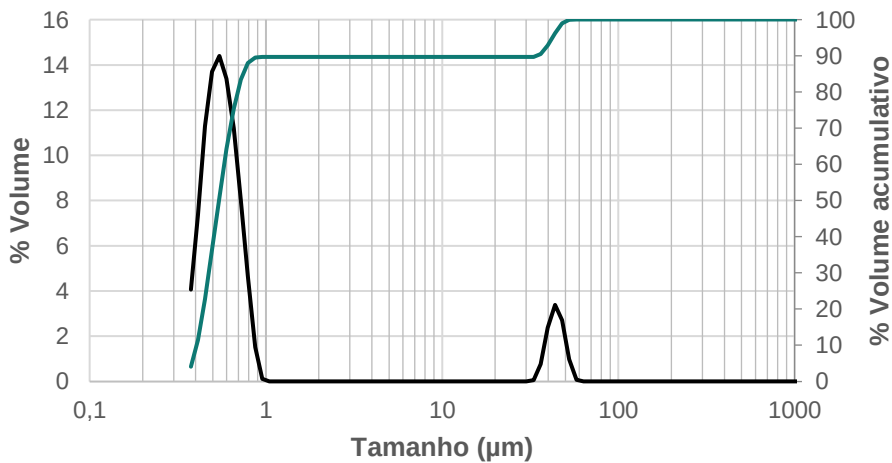


# 06 Anexos

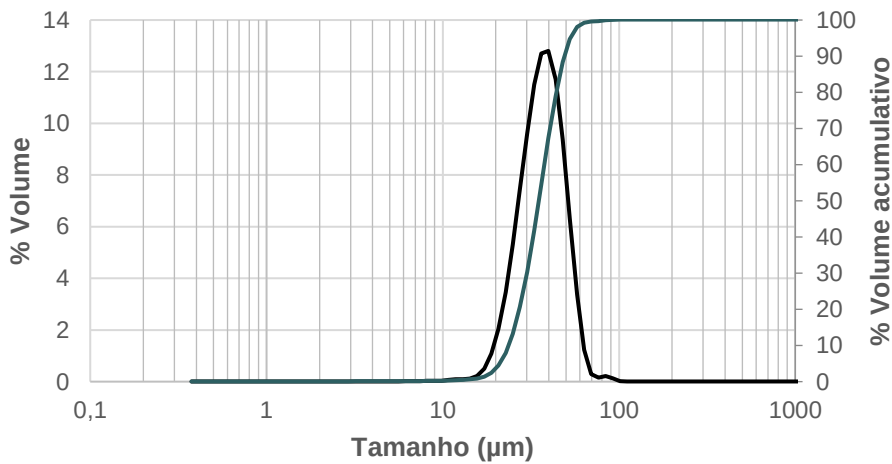
N – Pós novos de AISI 316L



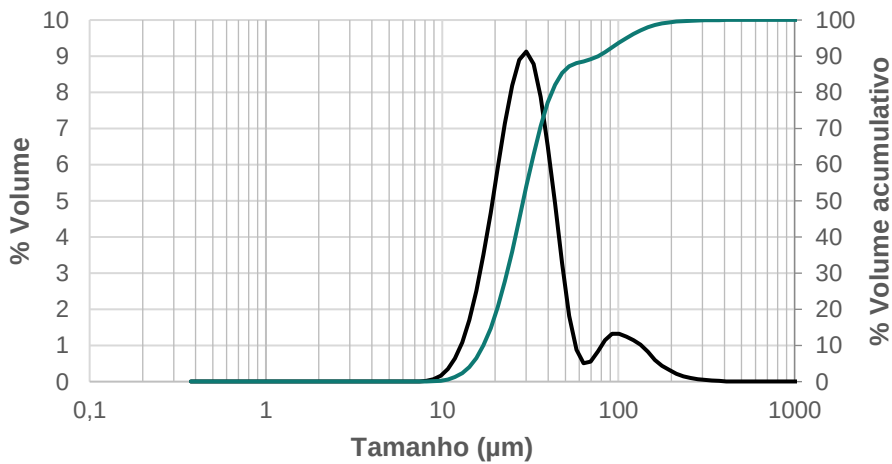
R – Pós recirculados 60x



A – Pós utilizados em 2 builds

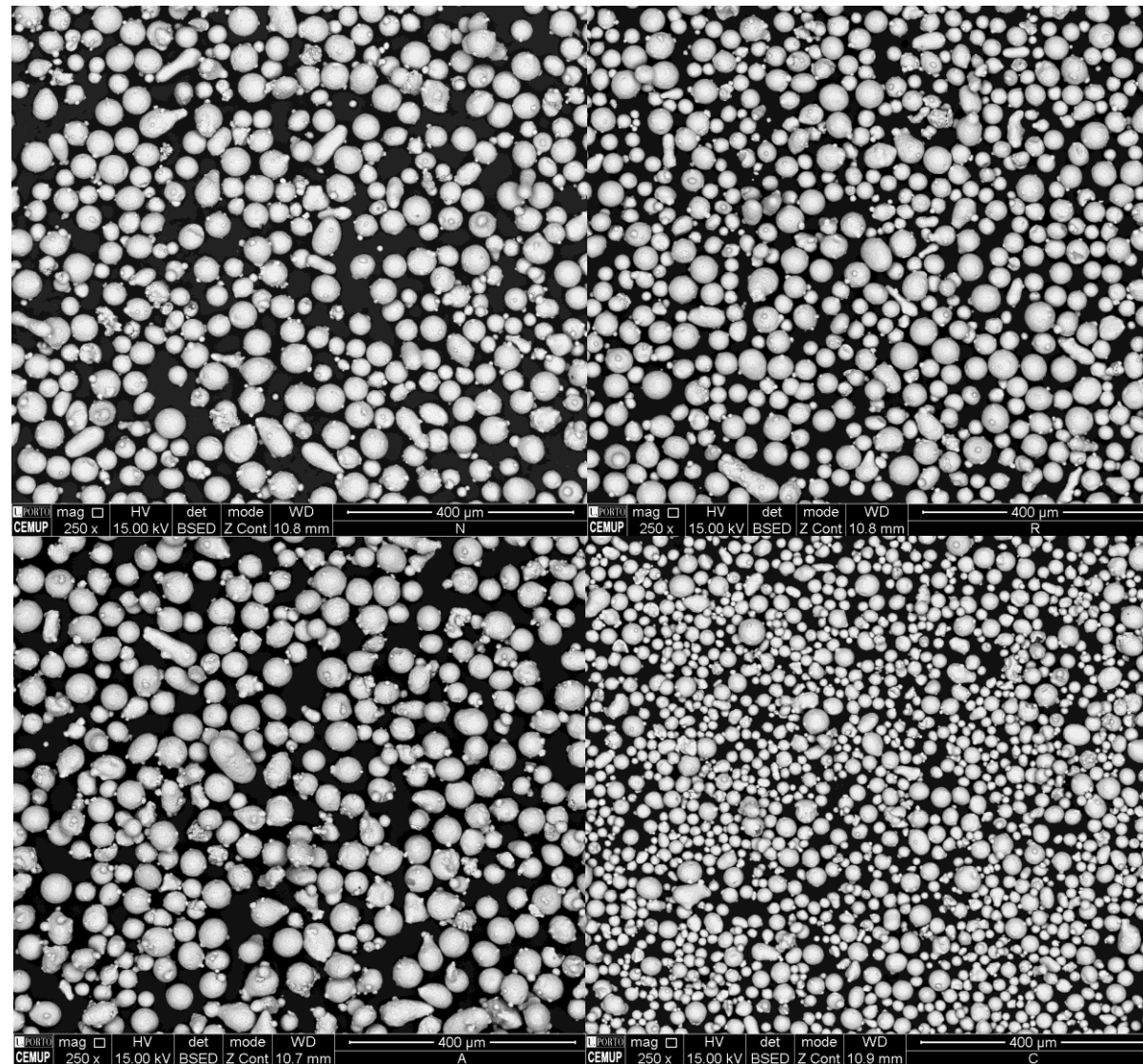


C – Pós rejeitados

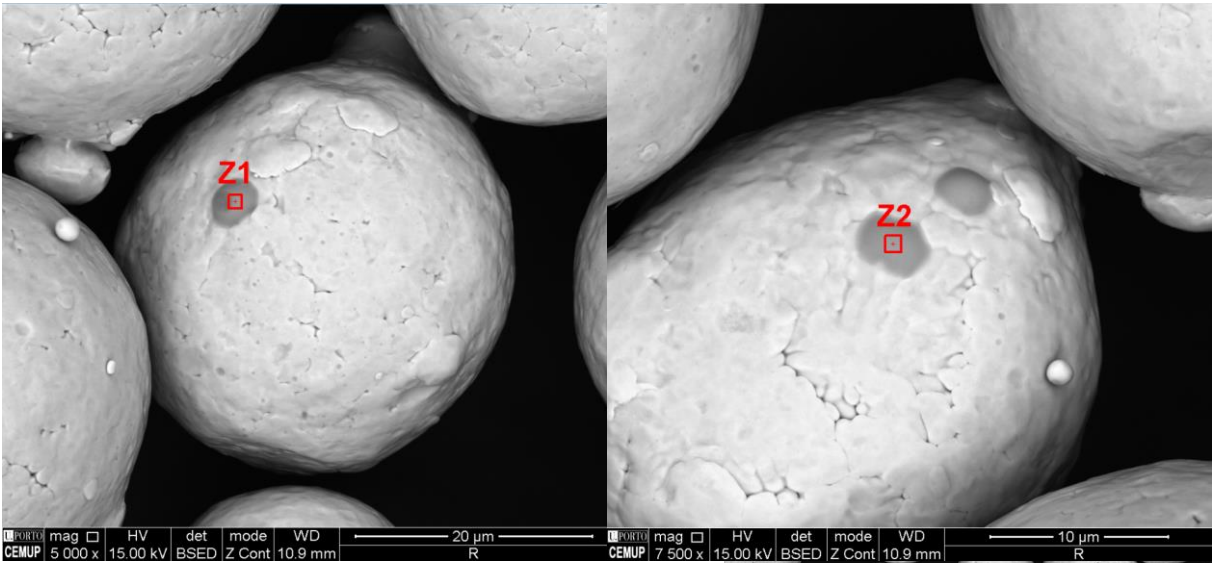


# Scanning electron microscopy (SEM)

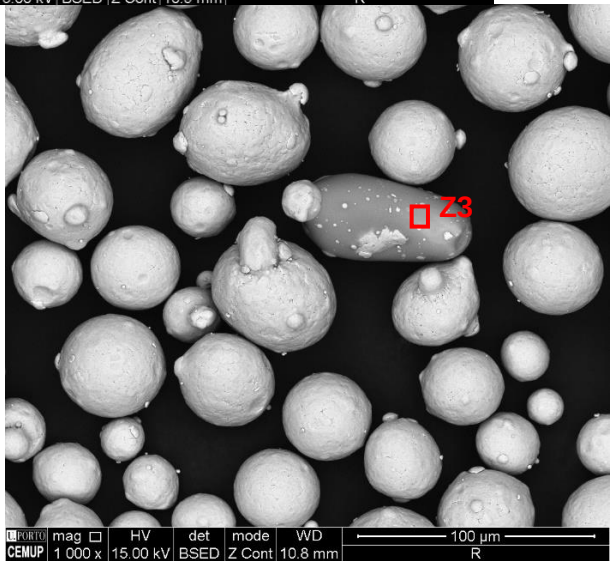
addcreative



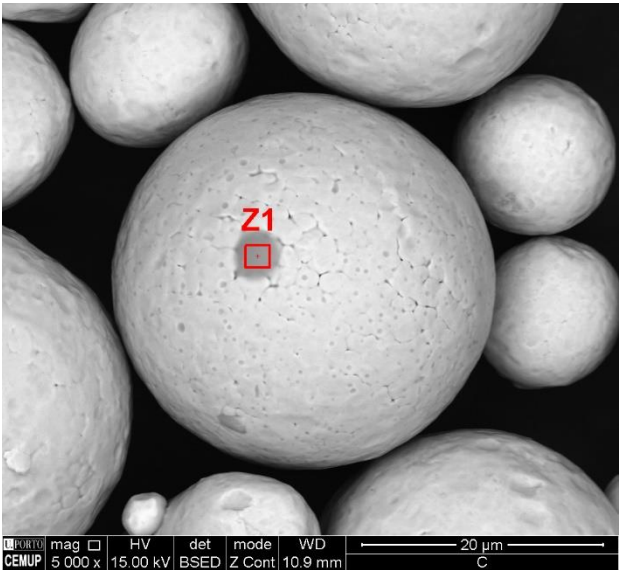
# Oxidação nos pós R



| Elemento | Z1    |       | Z2    |       | Z3    |       |
|----------|-------|-------|-------|-------|-------|-------|
|          | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   |
| Si       | 17.87 | 20.16 | 15.25 | 17.18 | 19.6  | 18.92 |
| Cr       | 13.18 | 8.03  | 11.84 | 7.2   | 10.6  | 5.53  |
| Mn       | 32.54 | 18.77 | 24.2  | 13.94 | 29.02 | 14.32 |
| Fe       | 9.87  | 5.6   | 20.73 | 11.75 | 4.24  | 2.06  |
| Ni       | 2.23  | 1.2   | 2.33  | 1.26  | -     | -     |
| O        | 20.31 | 40.24 | 21.94 | 43.41 | 30.91 | 52.36 |
| Al       | 3.12  | 3.67  | 3.09  | 3.63  | 4.7   | 4.72  |
| C        | 0.88  | 2.33  | 0.62  | 1.64  | 0.93  | 2.1   |



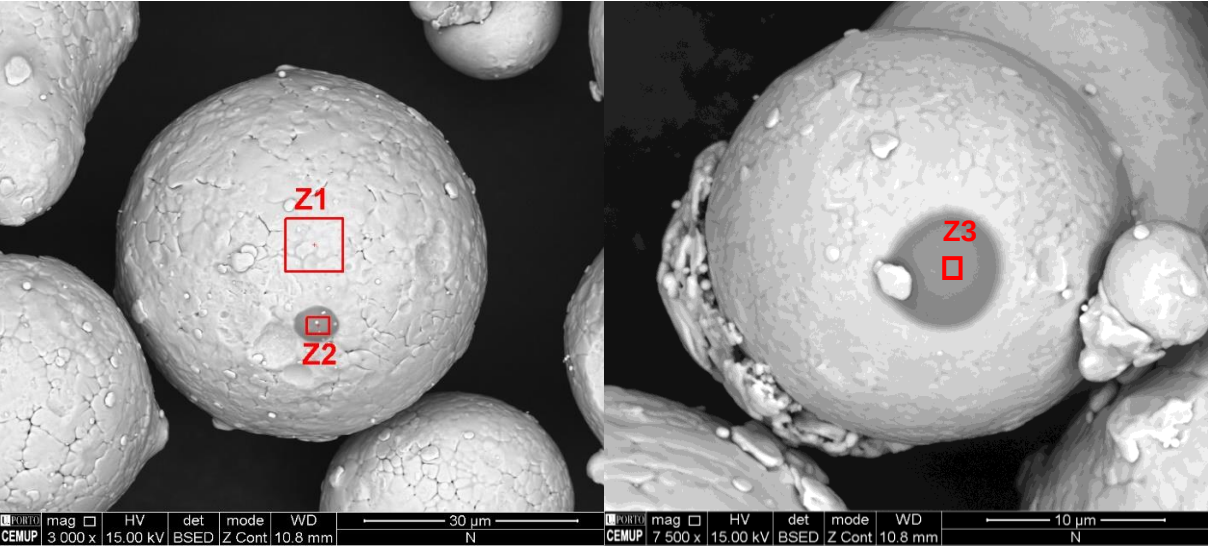
# Oxidação nos pós C



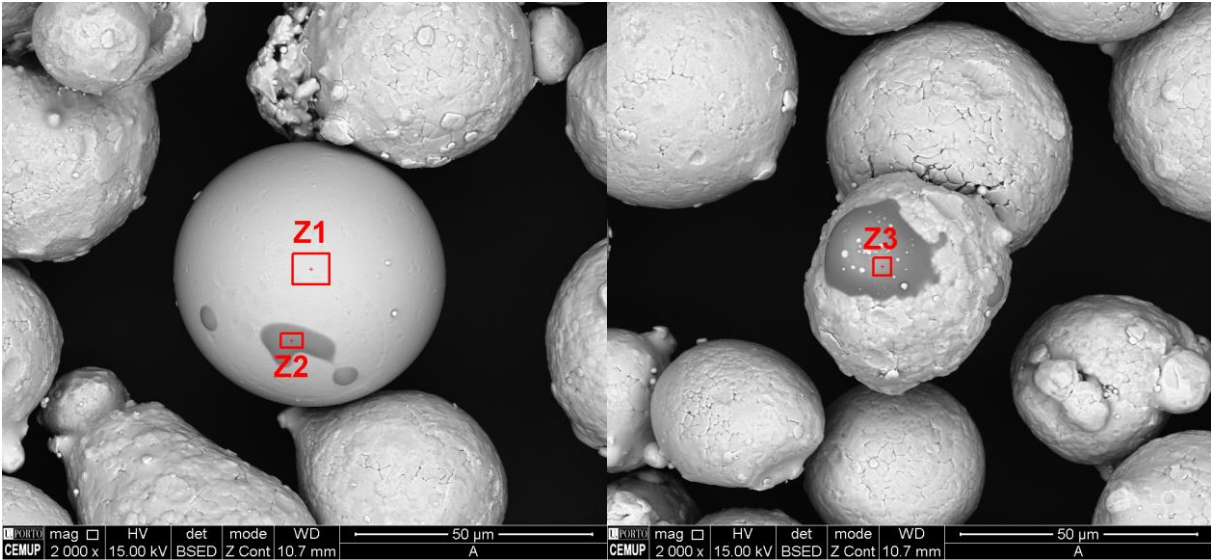
| Elemento | Z1    |       |
|----------|-------|-------|
|          | Wt%   | At%   |
| O        | 22.8  | 44.82 |
| Mg       | 0.56  | 0.72  |
| Al       | 1.27  | 1.48  |
| Si       | 17.34 | 19.42 |
| Mo       | 1     | 0.33  |
| Ca       | 1.28  | 1     |
| Ti       | 0.86  | 0.56  |
| Cr       | 14.14 | 8.55  |
| Mn       | 25    | 14.31 |
| Fe       | 13.26 | 7.47  |
| Ni       | 2.51  | 1.35  |



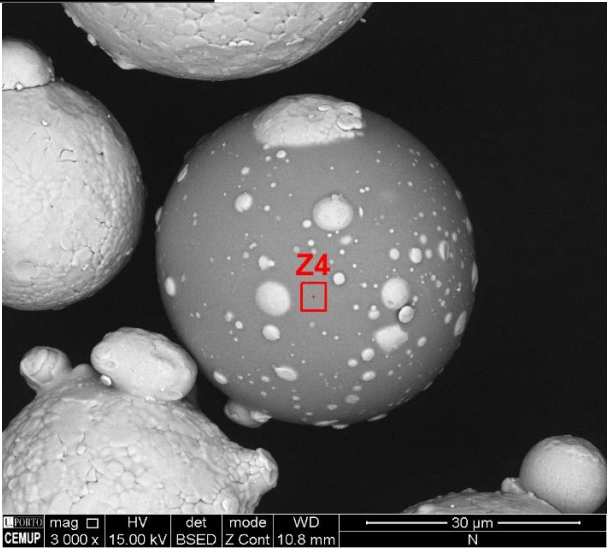
# Oxidação nos pós N



# Oxidação nos pós A



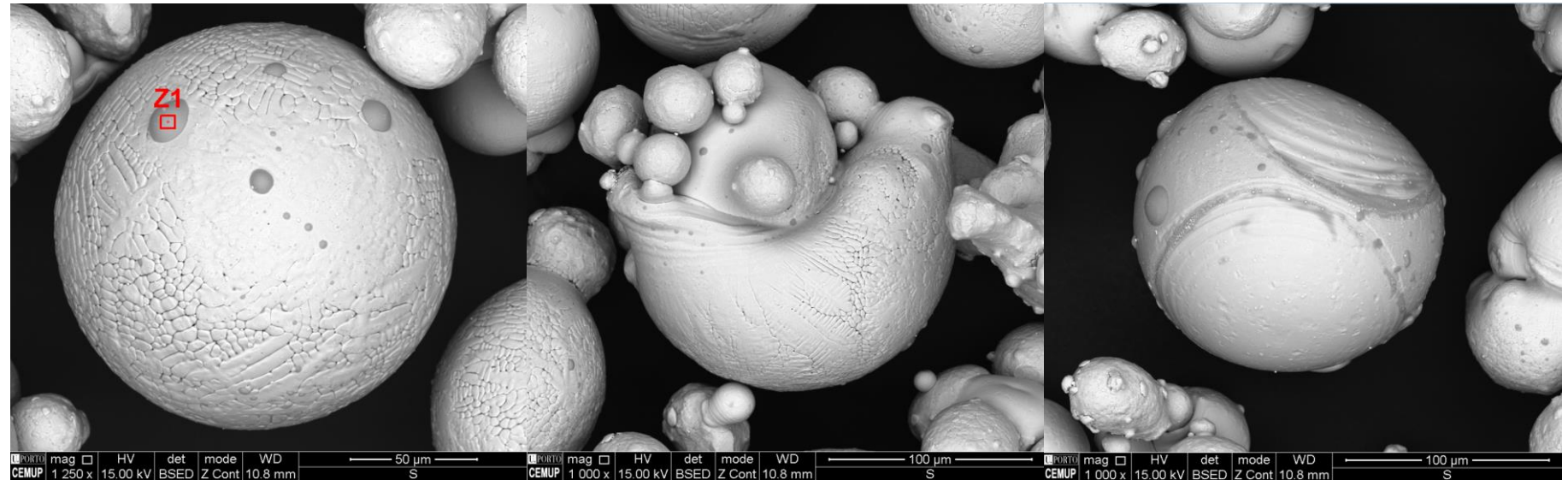
| Elemento | Z1    |       | Z2    |       | Z3    |       | Z4    |       |
|----------|-------|-------|-------|-------|-------|-------|-------|-------|
|          | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   |
| Si       | 1.02  | 2.03  | 19.08 | 19.65 | 16.36 | 15.92 | 15.09 | 19.16 |
| Mo       | 2.98  | 1.73  | 0.7   | 0.21  | -     | -     | -     | -     |
| Cr       | 16.77 | 17.92 | 10.15 | 5.65  | 9.17  | 4.82  | 15.94 | 10.93 |
| Mn       | 2.12  | 2.14  | 31.44 | 16.56 | 26.74 | 13.3  | 24.48 | 15.89 |
| Fe       | 65.22 | 64.92 | 6.1   | 3.16  | 4.22  | 2.06  | 23.84 | 15.22 |
| Ni       | 11.89 | 11.25 | -     | -     | -     | -     | 4.43  | 2.69  |
| O        | -     | -     | 27.04 | 48.89 | 28.54 | 48.74 | 16.21 | 36.12 |
| Al       | -     | -     | 5.48  | 5.87  | 14.98 | 15.16 | -     | -     |



| Elemento | Z1    |       | Z2    |       | Z3    |       |
|----------|-------|-------|-------|-------|-------|-------|
|          | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   |
| Si       | 0.66  | 1.28  | 18.83 | 23.53 | 21.5  | 20.71 |
| Mo       | 2.42  | 1.38  | -     | -     | -     | -     |
| Cr       | 16.26 | 17.92 | 9.02  | 6.09  | 12.48 | 6.49  |
| Mn       | -     | -     | 51.83 | 33.1  | 26.29 | 12.95 |
| Fe       | 69.34 | 67.73 | 4.64  | 2.92  | 2.68  | 1.3   |
| Ni       | 10.76 | 10    | -     | -     | -     | -     |
| O        | -     | -     | 15.67 | 34.36 | 31.01 | 52.43 |
| Al       | -     | -     | -     | -     | 14.98 | 15.16 |
| Mg       | -     | -     | -     | -     | 0.71  | 0.79  |
| C        | 0.56  | 2.55  | -     | -     | -     | -     |

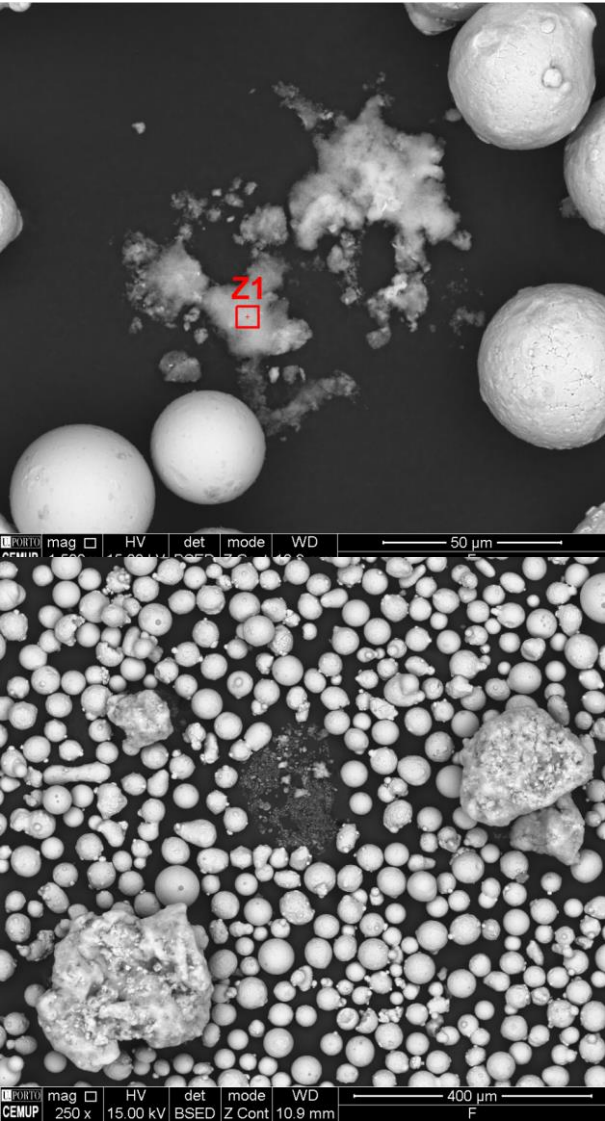
# SEM – Pós rejeitados pelo crivo superior (S)

| Elemento | Z1    |       |
|----------|-------|-------|
|          | Wt%   | At%   |
| C        | 1.03  | 2.37  |
| O        | 29.99 | 52.04 |
| Al       | 0.56  | 0.58  |
| Si       | 20.91 | 20.66 |
| Cr       | 12.92 | 6.9   |
| Mn       | 31.35 | 15.84 |
| Fe       | 3.24  | 1.61  |

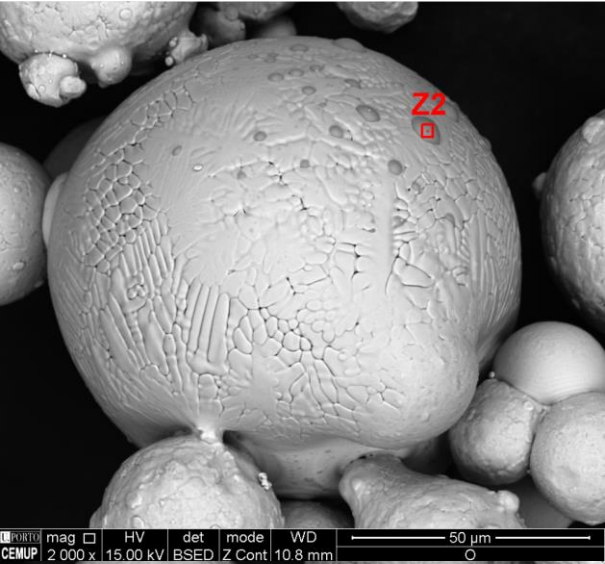
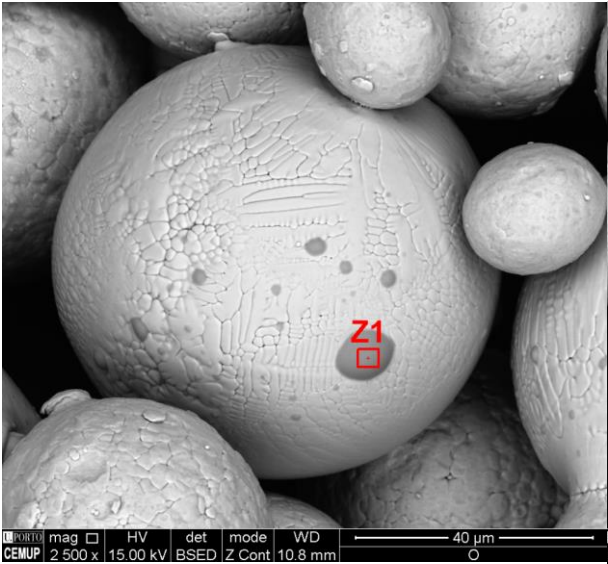




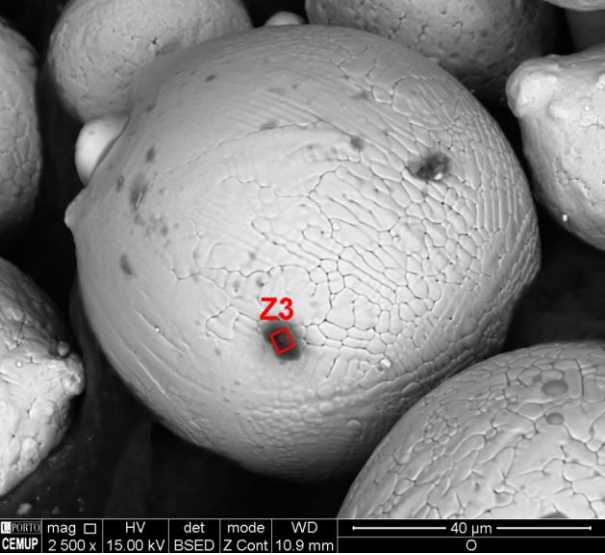
# SEM – Pós recolhidos pelo filtro (F)



| Elemento | Z1    |       |
|----------|-------|-------|
|          | Wt%   | At%   |
| C        | 15.58 | 36.5  |
| O        | 16.21 | 28.5  |
| Si       | 0.55  | 0.55  |
| Cr       | 12.67 | 6.85  |
| Mn       | 5.13  | 2.63  |
| Fe       | 43.36 | 21.85 |
| Ni       | 6.5   | 3.12  |

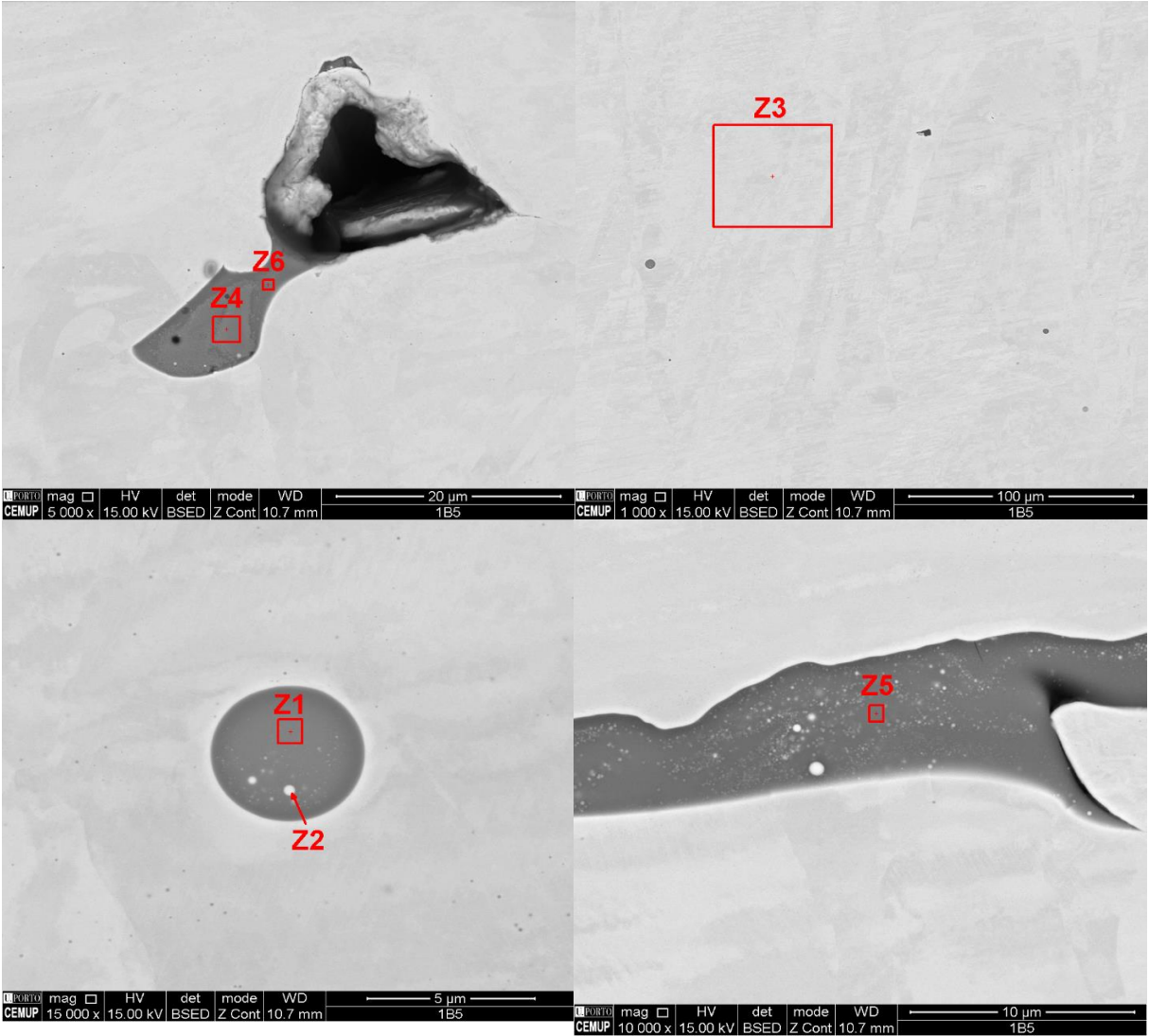
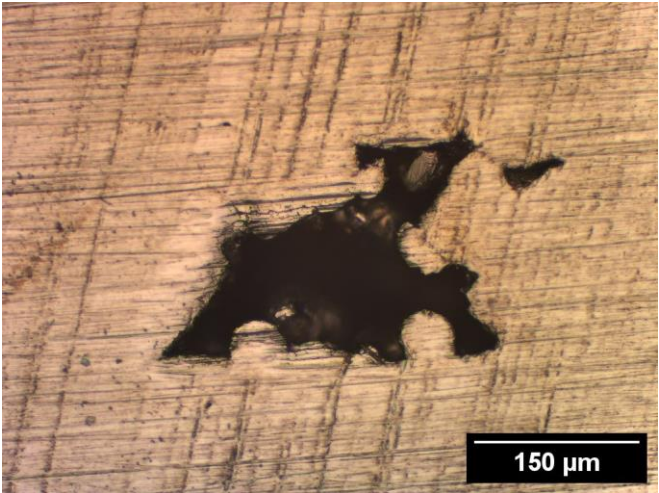


| Elemento | Z1    |       | Z2    |       | Z3    |      |
|----------|-------|-------|-------|-------|-------|------|
|          | Wt%   | At%   | Wt%   | At%   | Wt%   | At%  |
| C        | 0.82  | 2.01  | 1.11  | 2.51  | 69.43 | 83.6 |
| O        | 26.32 | 48.22 | 32.06 | 54.2  | 9.86  | 8.91 |
| Si       | 20.66 | 21.57 | 20.43 | 19.67 | 1.25  | 0.64 |
| Cr       | 12.4  | 6.99  | 10.04 | 5.22  | 1.91  | 0.53 |
| Mn       | 36.91 | 19.7  | 32.2  | 15.85 | -     | -    |
| Fe       | 2.88  | 1.51  | 3.11  | 1.51  | 5.17  | 1.34 |
| Al       | -     | -     | 1.04  | 1.04  | -     | -    |
| Na       | -     | -     | -     | -     | 2.16  | 1.36 |
| S        | -     | -     | -     | -     | 3.13  | 1.41 |
| Cl       | -     | -     | -     | -     | 1.01  | 0.41 |
| K        | -     | -     | -     | -     | 1.89  | 0.7  |
| Ca       | -     | -     | -     | -     | 1.14  | 0.41 |
| Zn       | -     | -     | -     | -     | 3.04  | 0.67 |



# Build 1B5 – SEM & OM

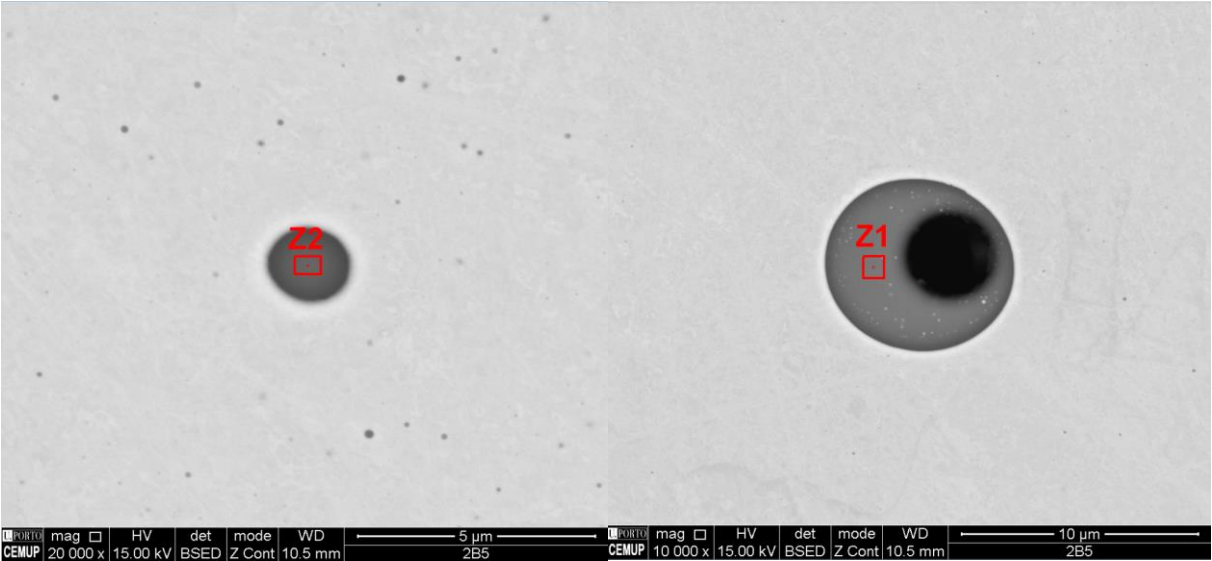
| Elemento | Z1    |       | Z2    |       | Z3    |       | Z4    |       | Z5    |       | Z6    |       |
|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|          | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   |
| O        | 24.44 | 46.72 | 14.47 | 34.07 | -     | -     | 27.46 | 47.82 | 27.05 | 48.02 | 24.66 | 45.44 |
| Si       | 20.1  | 21.89 | 11.28 | 15.13 | 0.85  | 1.69  | 9.65  | 9.57  | 20.00 | 20.23 | 13.33 | 13.99 |
| Cr       | 16.52 | 9.72  | 15.01 | 10.87 | 16.59 | 17.74 | 11.76 | 6.3   | 13.19 | 7.21  | 12.39 | 7.03  |
| Mn       | 38.95 | 21.58 | 15.68 | 10.75 | 1.79  | 1.81  | 24.75 | 12.55 | 32.96 | 17.04 | 24.67 | 13.24 |
| Fe       | -     | -     | 37.47 | 25.27 | 66.89 | 66.57 | 6.51  | 3.25  | -     | -     | 10.69 | 5.64  |
| Ni       | -     | -     | 6.09  | 3.91  | 11.32 | 10.71 | -     | -     | -     | -     | 1.59  | 0.8   |
| Mo       | -     | -     | -     | -     | 2.52  | 1.48  | -     | -     | -     | -     | -     | -     |
| Al       | -     | -     | -     | -     | -     | -     | 19.87 | 20.52 | 3.73  | 3.93  | 12.68 | 13.86 |
| Mg       | -     | -     | -     | -     | -     | -     | -     | -     | 3.06  | 3.58  | -     | -     |





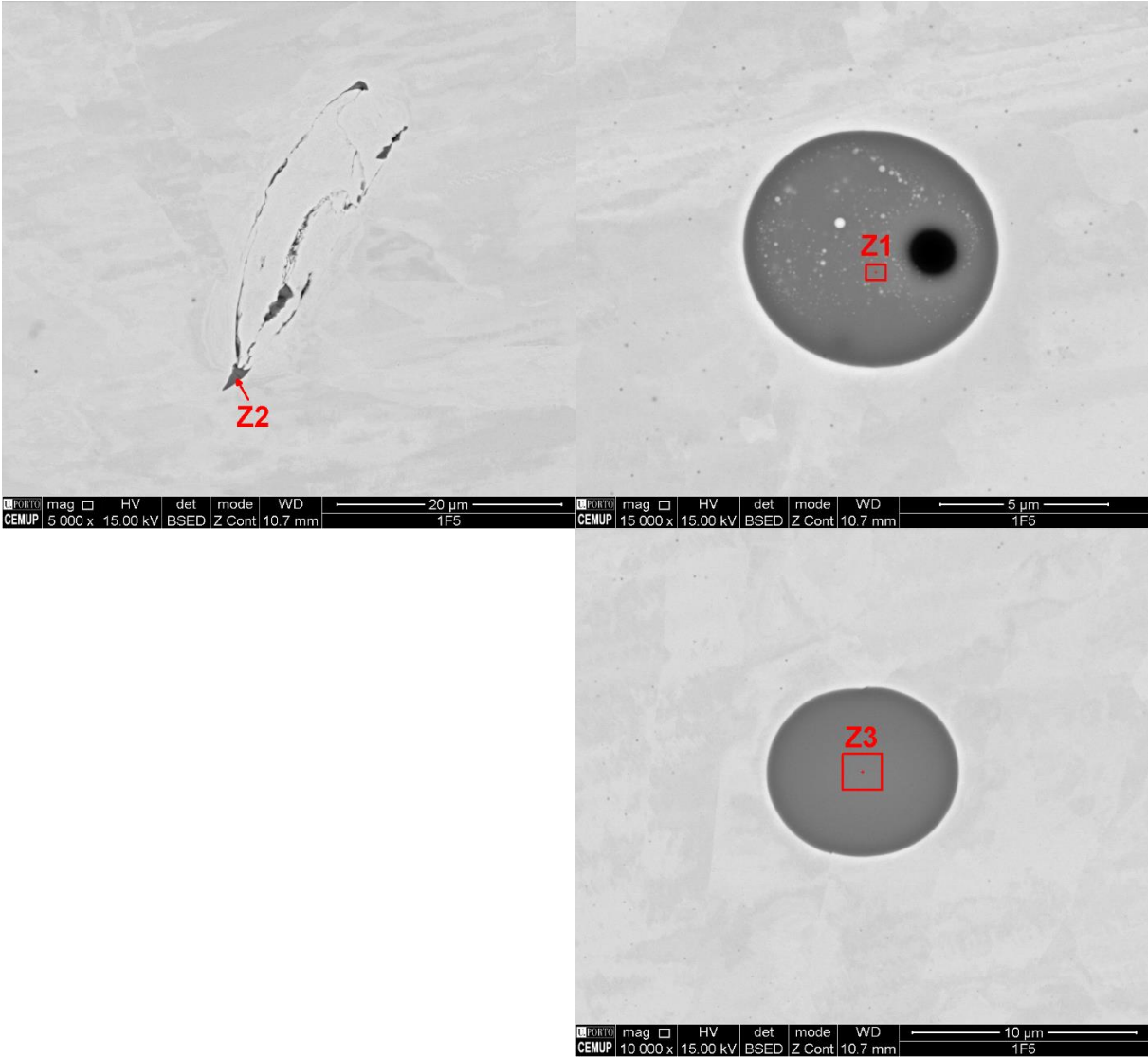
# Build 2B5 – SEM

| Elemento | Z1    |       | Z2    |       |
|----------|-------|-------|-------|-------|
|          | Wt%   | At%   | Wt%   | At%   |
| O        | 23.06 | 44.88 | 21.35 | 42.43 |
| Al       | 2.06  | 2.38  | 3.39  | 4     |
| Si       | 18.13 | 20.1  | 17.15 | 19.42 |
| Cr       | 17.33 | 10.38 | 14.85 | 9.08  |
| Mn       | 31.38 | 17.78 | 34.84 | 20.17 |
| Fe       | 8.05  | 4.49  | 7.41  | 4.22  |
| Ti       | -     | -     | 1.02  | 0.68  |



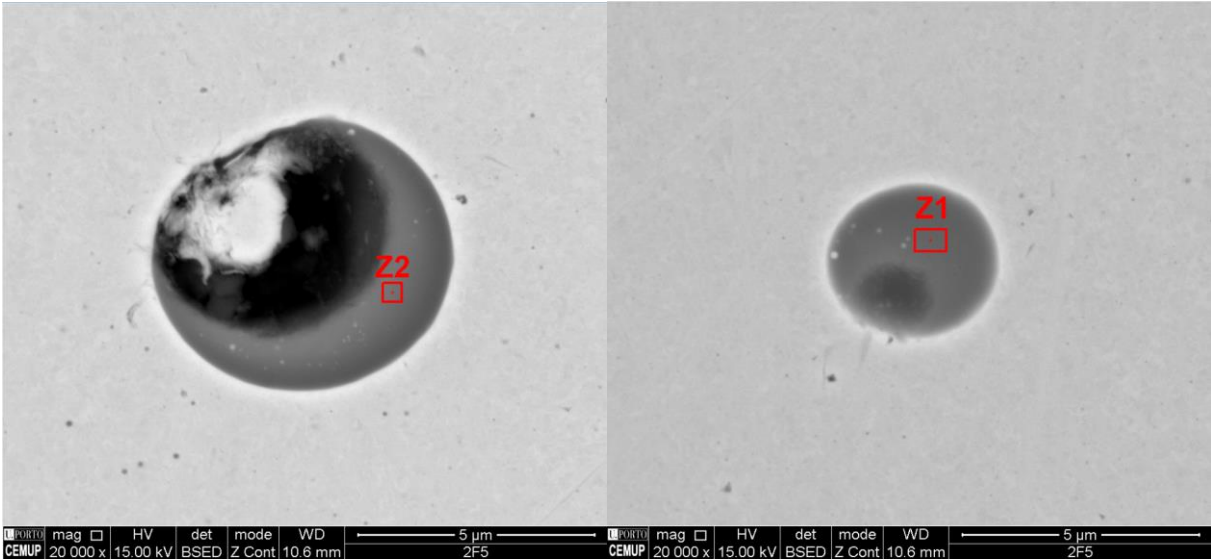
# Build 1F5 – SEM

| Elemento | Z1    |       | Z2    |       | Z3    |       |
|----------|-------|-------|-------|-------|-------|-------|
|          | Wt%   | At%   | Wt%   | At%   | Wt%   | At%   |
| O        | 27.06 | 49.31 | 15.42 | 35.08 | 28.6  | 49.98 |
| Al       | 3.48  | 3.76  | 1.73  | 2.33  | 6.51  | 6.75  |
| Si       | 18.81 | 19.53 | 12.47 | 16.16 | 20.45 | 20.36 |
| Cr       | 17.49 | 9.81  | 16.22 | 11.35 | 10.73 | 5.77  |
| Mn       | 33.16 | 17.6  | 24.09 | 15.96 | 31.46 | 16.01 |
| Fe       | -     | -     | 24.71 | 16.11 | 2.26  | 1.13  |
| Ni       | -     | -     | 4.03  | 2.5   | -     | -     |
| Mo       | -     | -     | 1.33  | 0.51  | -     | -     |

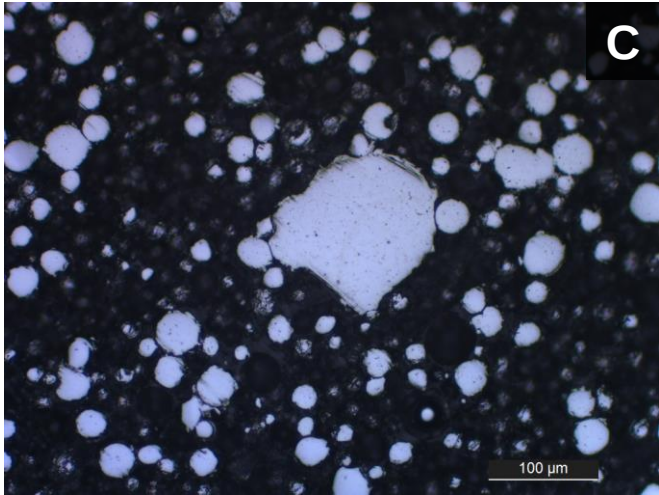
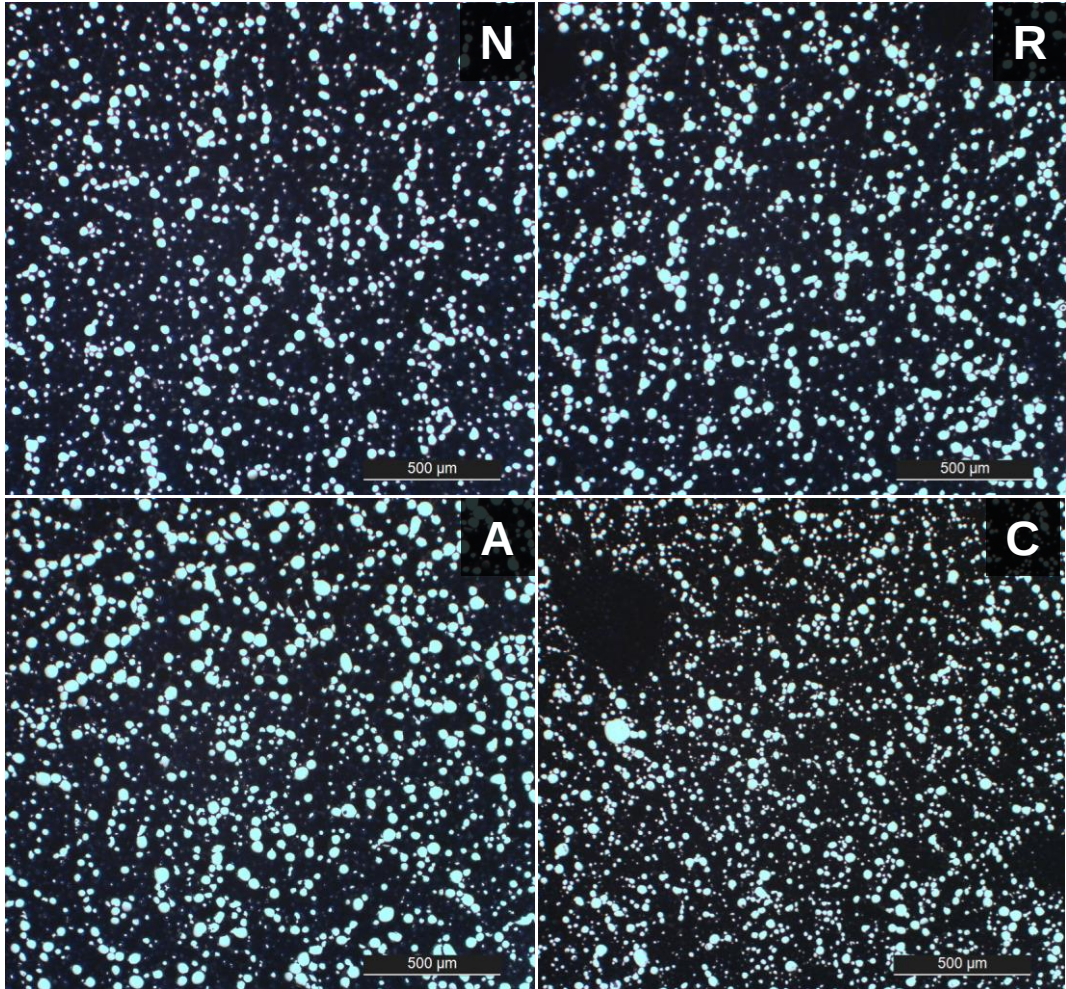


# Build 2F5 – SEM

| Elemento | Z1    |       | Z2    |       |
|----------|-------|-------|-------|-------|
|          | Wt%   | At%   | Wt%   | At%   |
| O        | 21.45 | 43.56 | 23.86 | 43.2  |
| Al       | 0.74  | 0.89  | 1.02  | 1.1   |
| Si       | 16.33 | 18.9  | 18.23 | 18.8  |
| Mo       | 0.59  | 0.2   | -     | -     |
| Cr       | 18.19 | 11.37 | 14.51 | 8.09  |
| Mn       | 29.76 | 17.6  | 34.58 | 18.23 |
| Fe       | 11.32 | 6.58  | 4.36  | 2.26  |
| Ni       | 1.63  | 0.9   | -     | -     |
| C        | -     | -     | 3.45  | 8.32  |



# OM – powder profile







Rua das Lages 67, 4410-272

Vila Nova de Gaia Portugal

GPS\_41°05'42.6"N 8°36'28.3"W

**T.** +351 226 192 700

**E.** [info@addcreative.tech](mailto:info@addcreative.tech)

[www.addcreative.tech](http://www.addcreative.tech)

**FB.** @Addcreative

**IG.** @addcreative

**in.** Addcreative